

PUBLISHED: 15-AUG-2017
2015.0 RANGE ROVER (LG), 303-01

ENGINE - TDV8 4.4L DIESEL

TIMING DRIVE COMPONENTS (G1509180)

REMOVAL AND INSTALLATION

12.65.13	TIMING CHAIN, GEARS AND TENSIONER - RENEW	LEFT HAND DRIVE, 4400 CC, TDV8, WITH PARTICULATE FILTER, WITHOUT DIESEL EXHAUST FLUID, WITHOUT FUEL FI	19.5	USED WITHINS	
12.65.13	TIMING CHAIN, GEARS AND TENSIONER - RENEW	LEFT HAND DRIVE, 4400 CC, TDV8, WITH PARTICULATE FILTER, WITHOUT FUEL FIRED BOOSTER HEATER	19.4	USED WITHINS	

SPECIAL TOOL(S)



E131271

JLR-303-1568

Locking Tool,
Camshaft



E148198

JLR-303-1620

Timing Pin



E131589

JLR-310-246

Locking Tool, Fuel Injection Pump



E131588

JLR-310-247

Locking Tool, Camshaft



PART(S)

STEP	REPLACE PART / RENEW PART	PART NAME	
Installation Step 14	Renew Part	Camshaft sprocket bolt	4
Installation Step 57	Renew Part	Fuel pump O-ring seal	1
Installation Step 59	Renew Part	Fuel Injection Pump retaining bolts	3

REMOVAL

 **CAUTION:**

Make sure that all openings are sealed. Use new blanking caps.

 **NOTES:**

- Some variation in the illustrations may occur, but the essential information is always correct.
- Removal steps in this procedure may contain installation details.
- Engine shown removed for clarity.

1. Disconnect the battery ground cable.

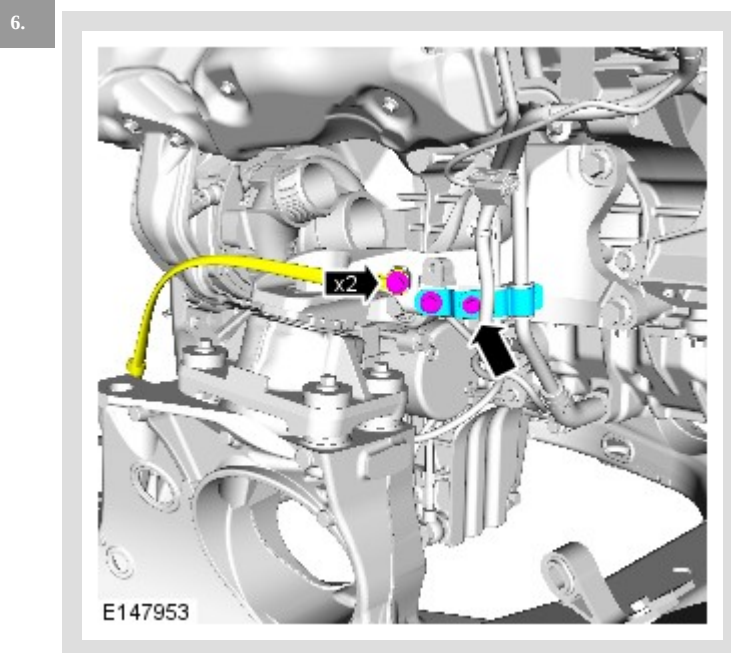
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).

2. Refer to: [Right Valve Cover](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).

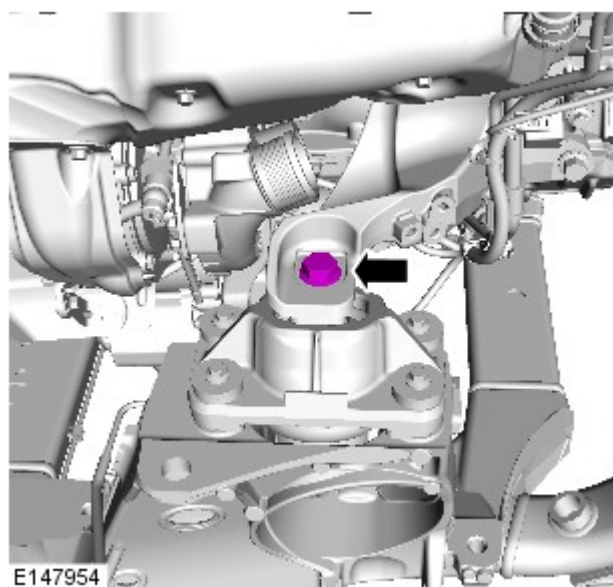
3. Refer to: [Left Valve Cover](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).

4. Refer to: [Generator](#) (414-02B Generator and Regulator - TDV8 4.4L Diesel, Removal and Installation).

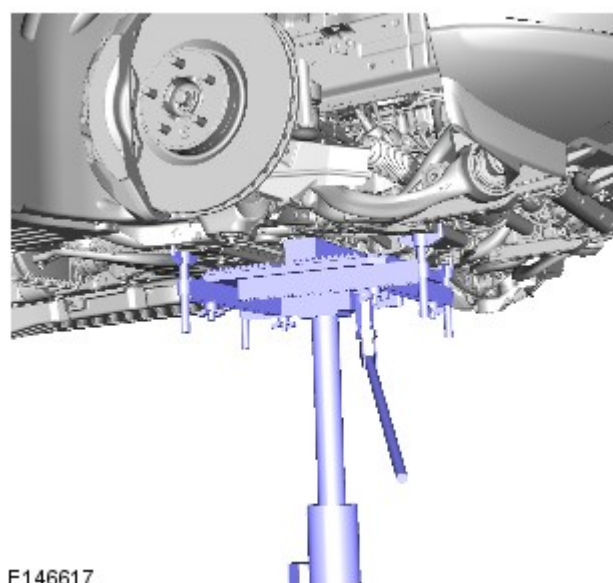
5. Refer to: [Brake Vacuum Pump - TDV8 4.4L Diesel](#) (206-07 Power Brake Actuation, Removal and Installation).



7.



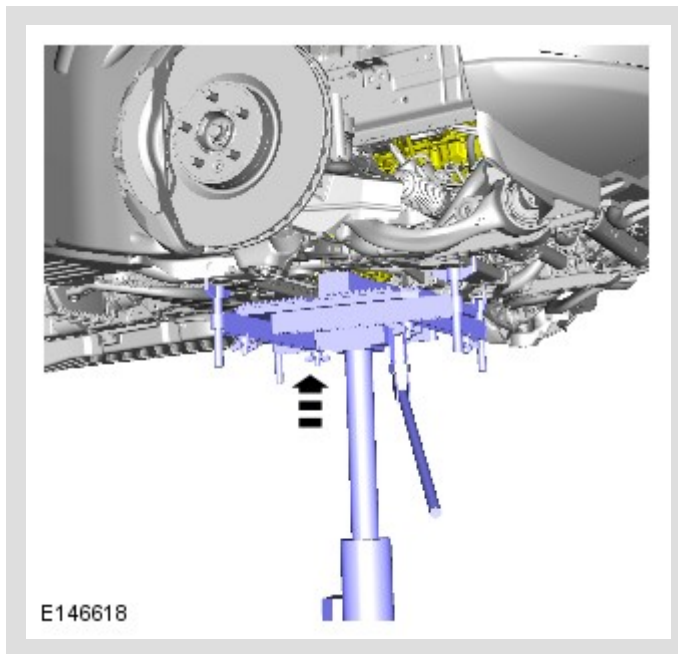
8.



9.

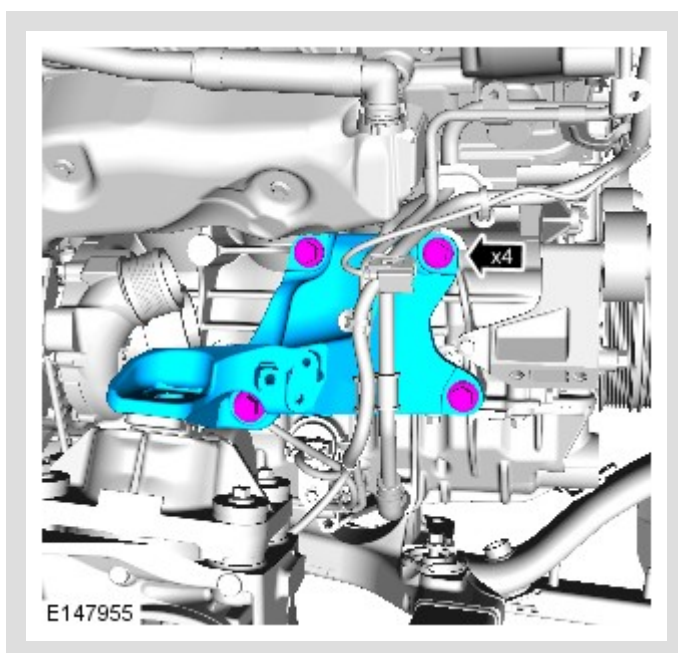
⚠ CAUTION:

Use a wooden block to protect the oil pan when supporting the engine.

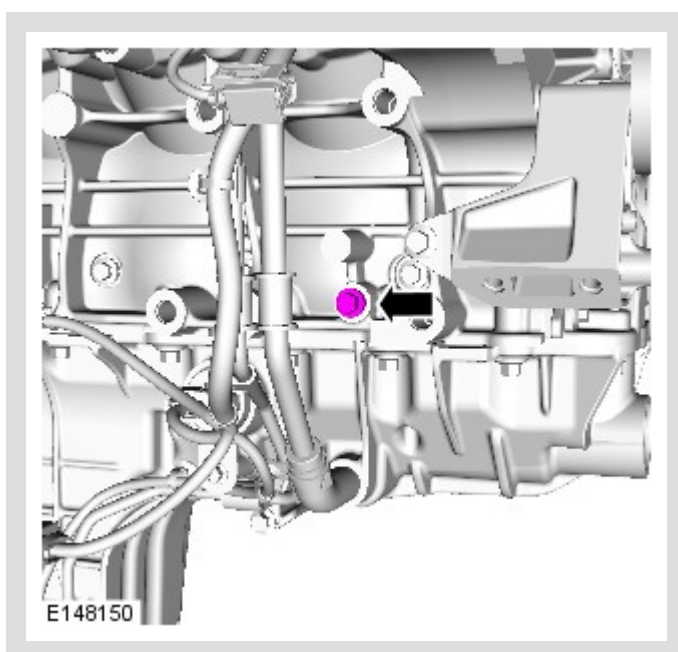


Using a suitable hydraulic jack, raise and support the engine.

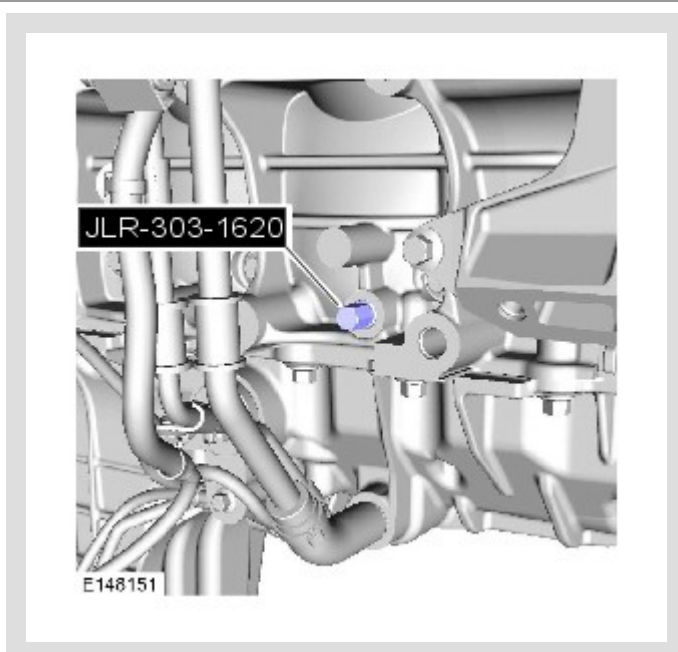
10.



11.

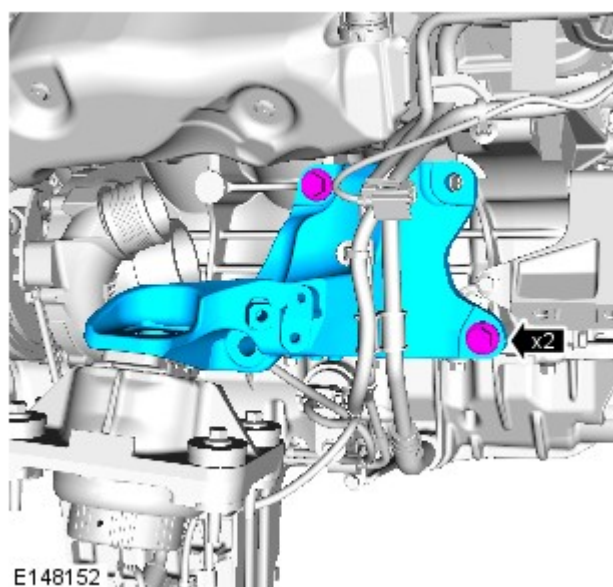


12.

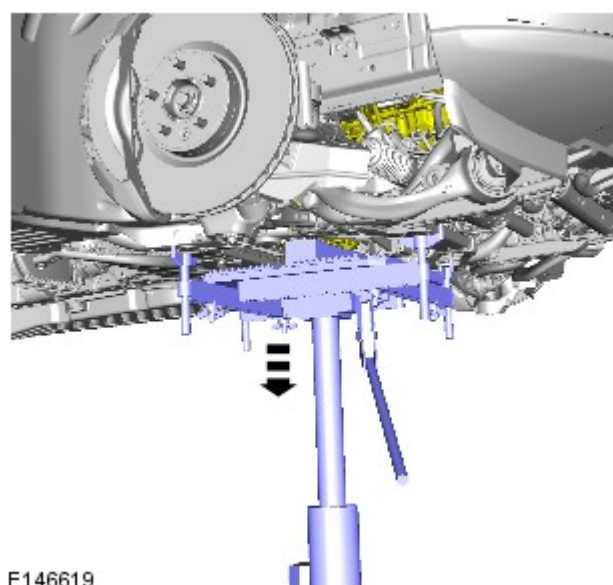


Special Tool(s): [JLR-303-1620](#)

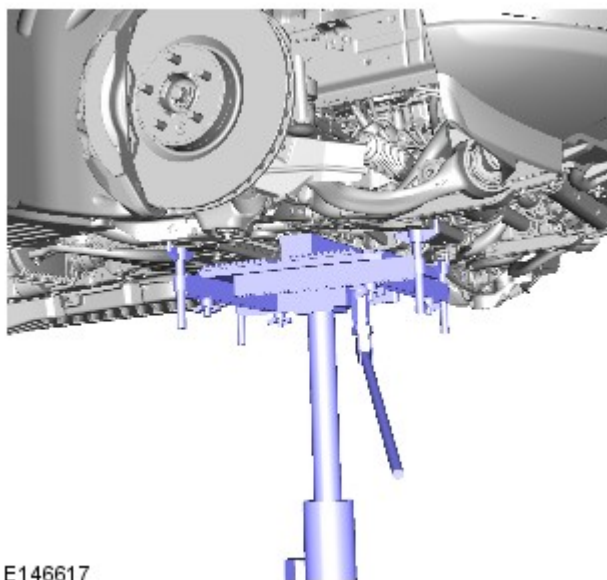
13.



14.

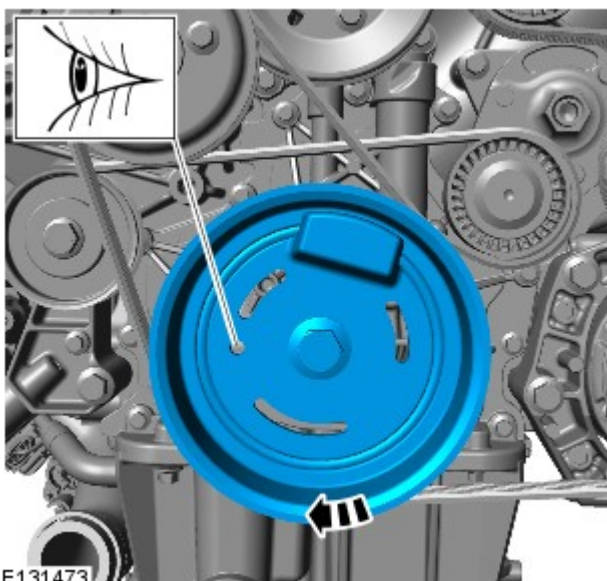


15.



E146617

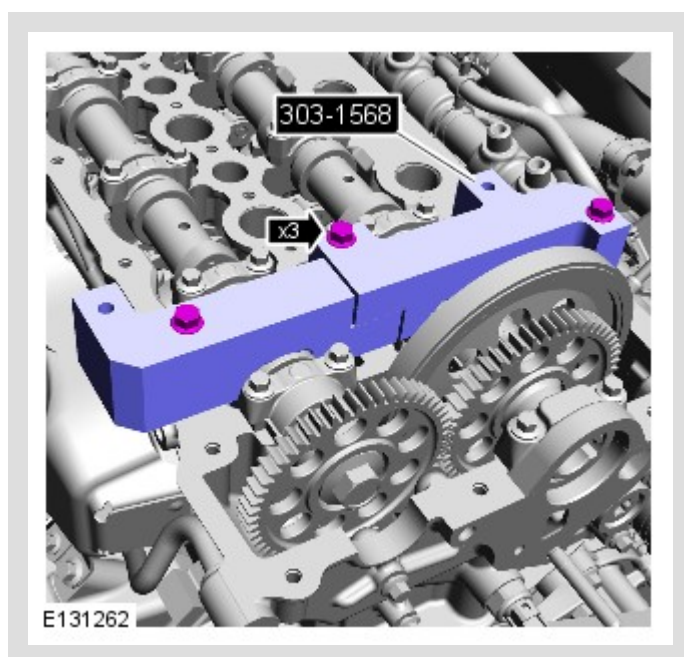
16.



E131473

Rotate the engine clockwise until the crankshaft contacts the timing pin. Do not attempt to rotate the engine counter clockwise.

17.

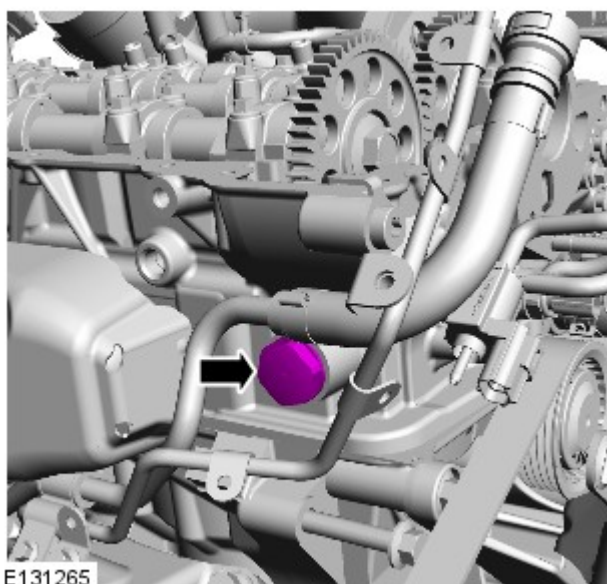


Special Tool(s): [JLR-303-1568](#)

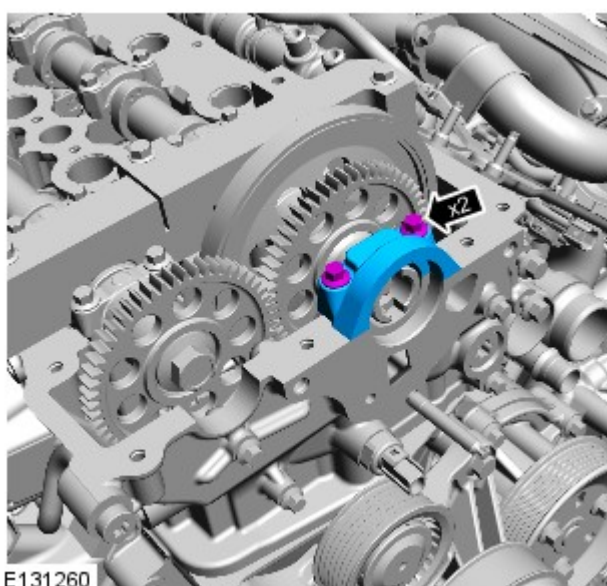
18.

ⓘ CAUTION:

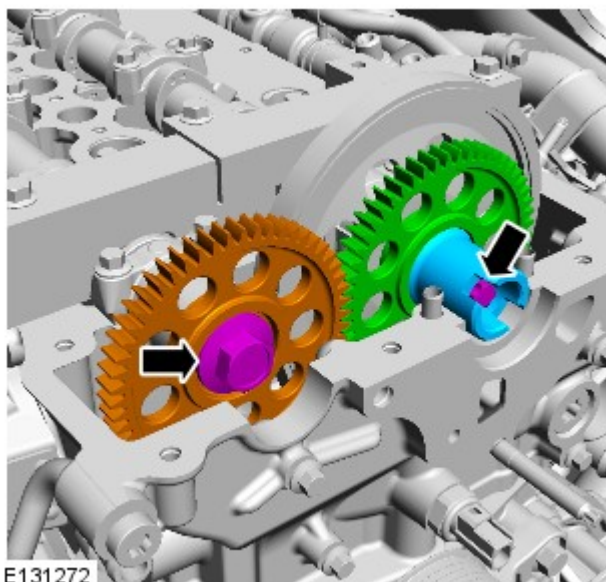
Discard the component.



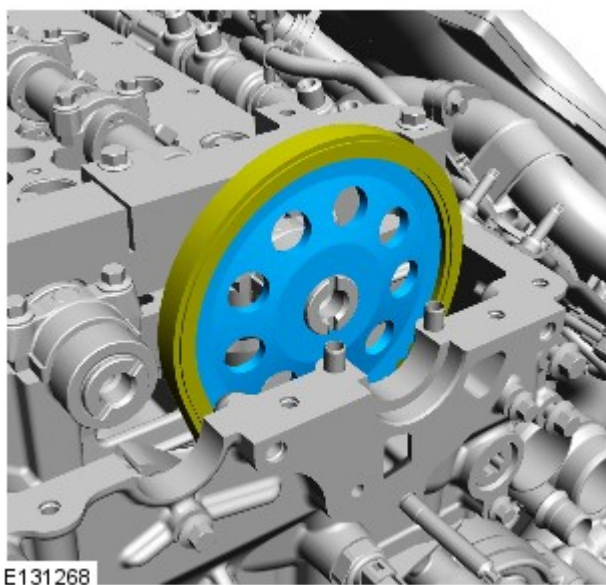
19.



20.



21.



22.

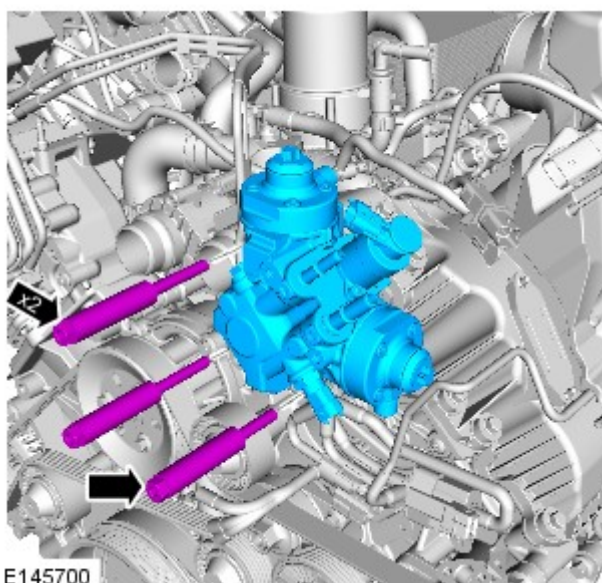
⚠ CAUTION:

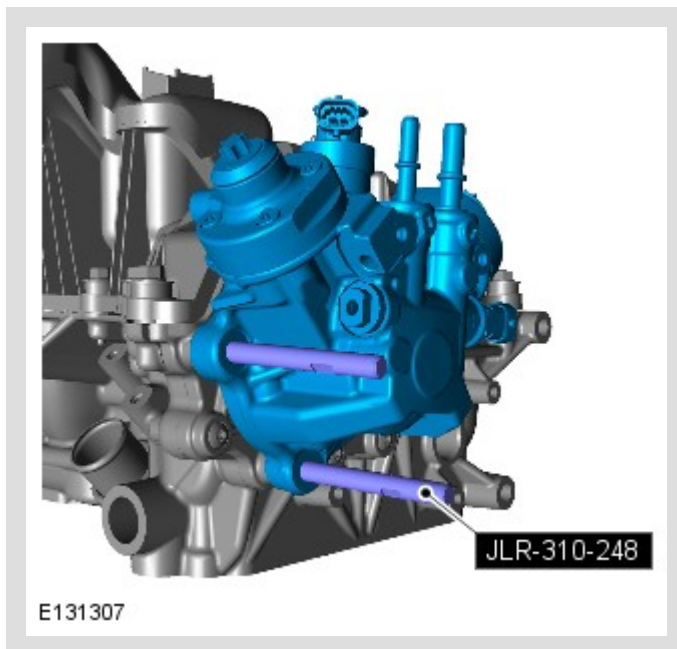
Discard the bolts.



NOTE:

Note the different lengths of the bolts.

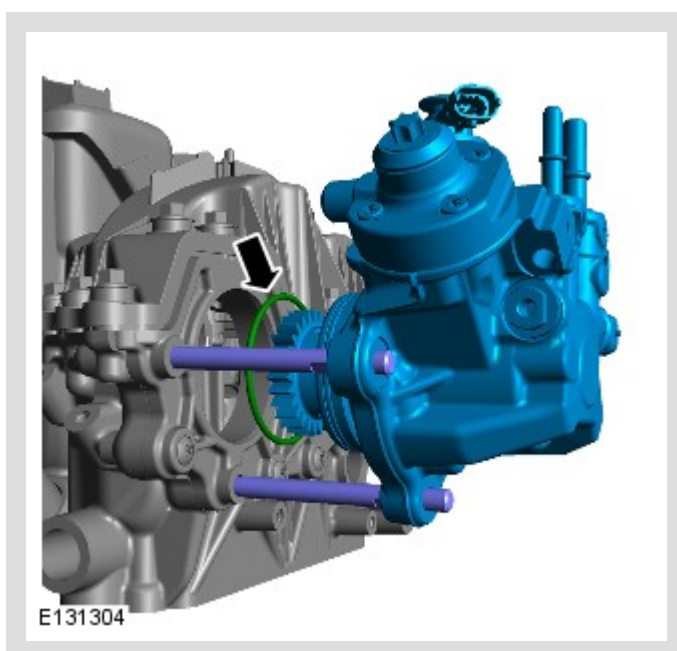




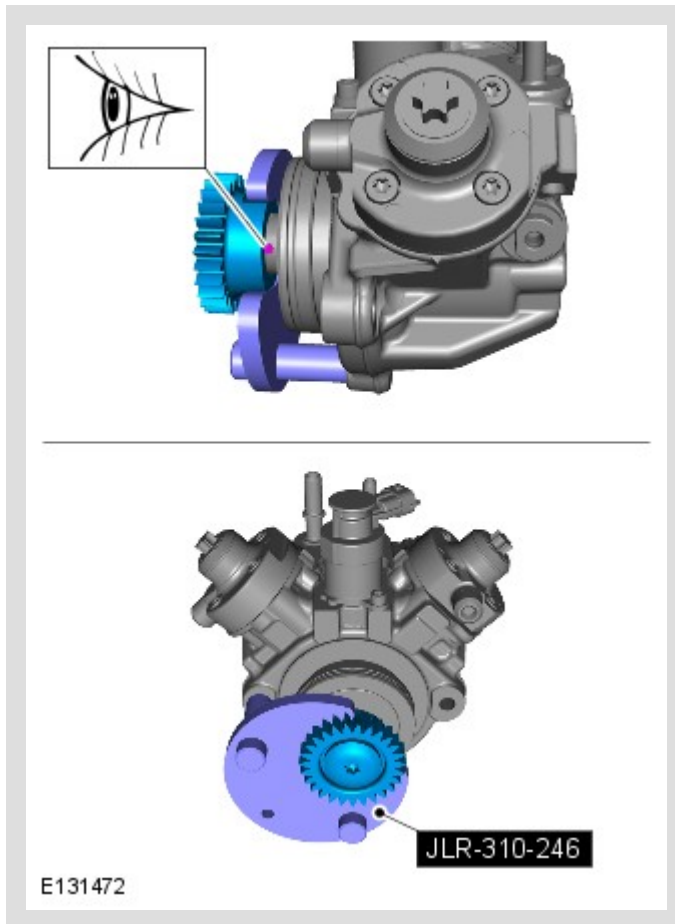
Install 2 guide pins in the original location of the 2 bolts.

Special Tool(s): [JLR-310-248](#)

24.

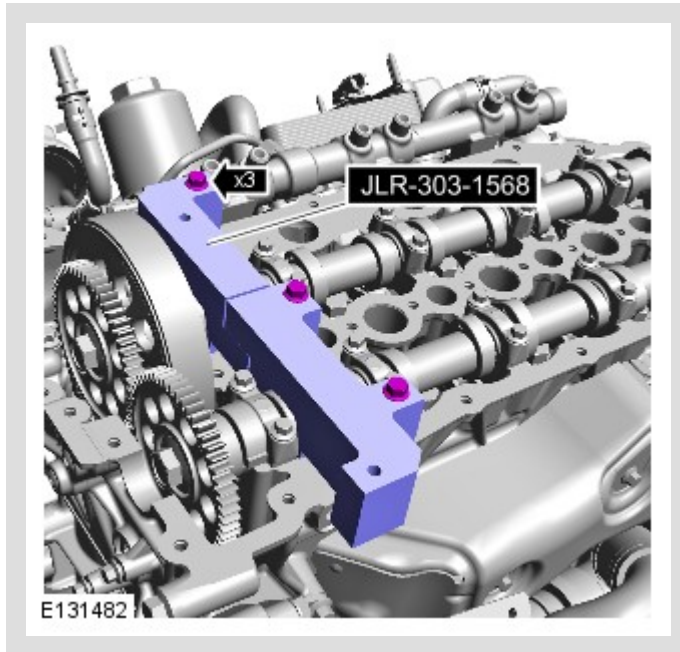


25.



Special Tool(s): [JLR-310-246](#)

26.

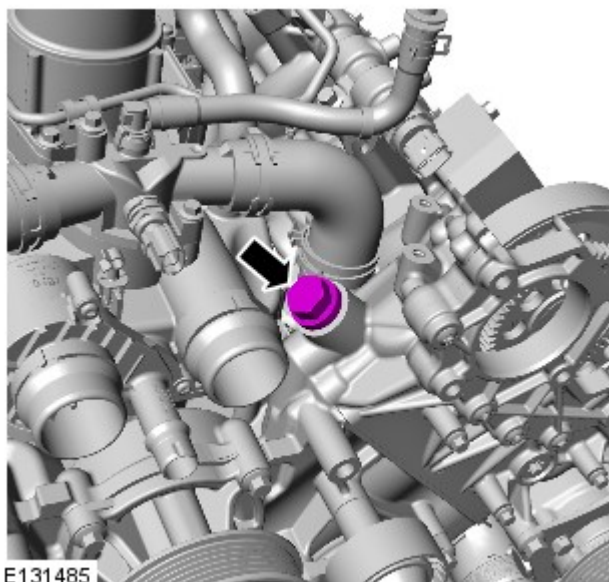


Special Tool(s): [JLR-303-1568](#)

27.

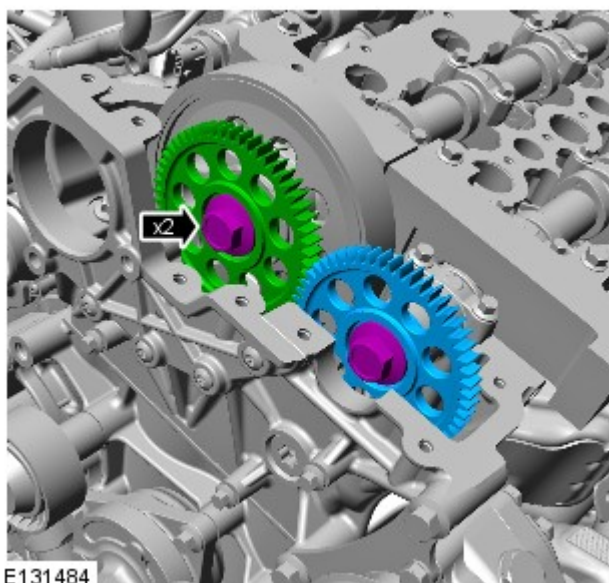
ⓘ CAUTION:

Discard the component.



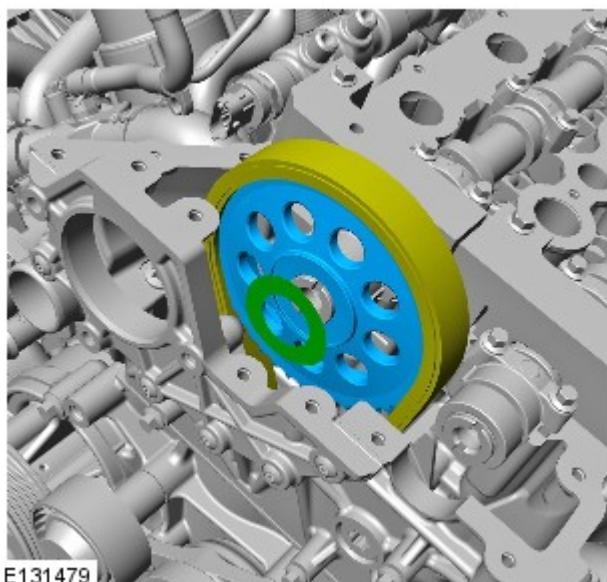
E131485

28.



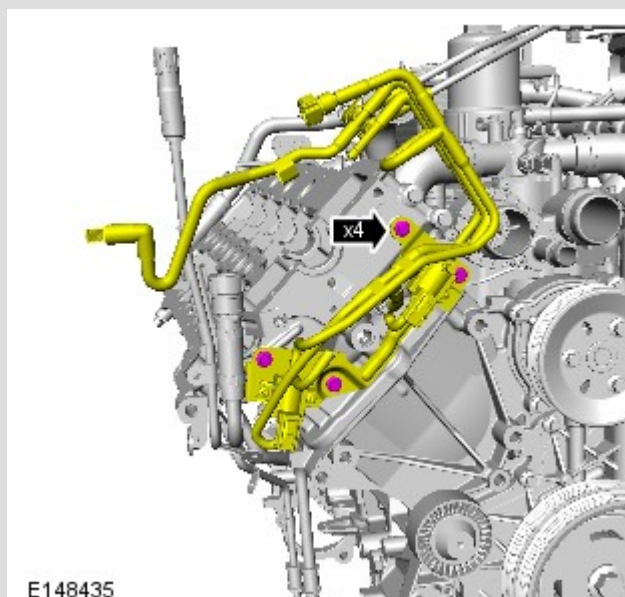
E131484

29.

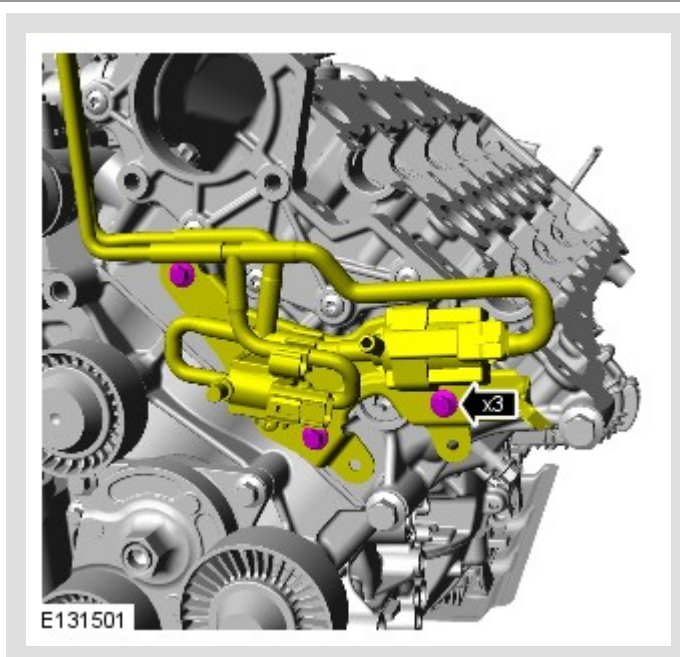


30. Refer to: [Oil Pump](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).

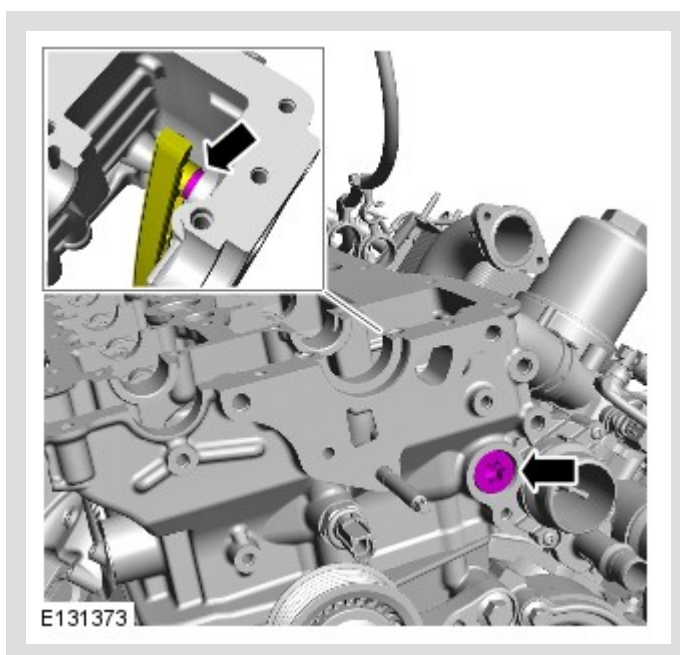
31.



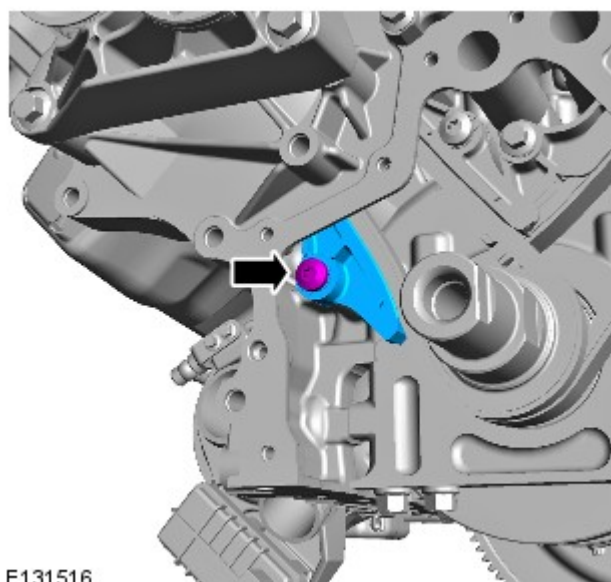
32.



33.

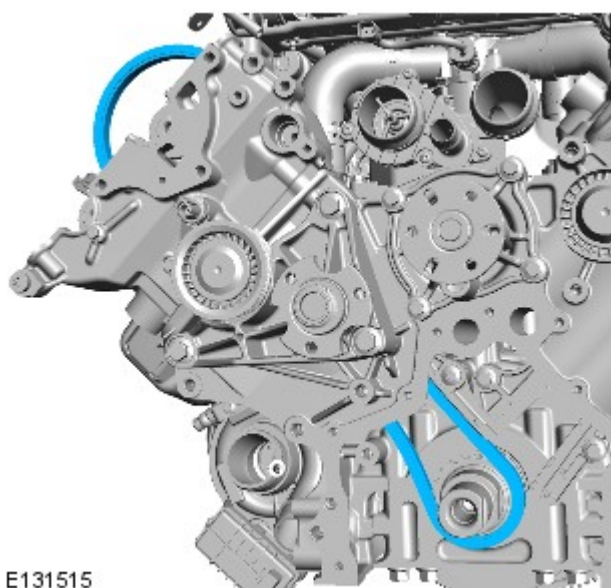


34.



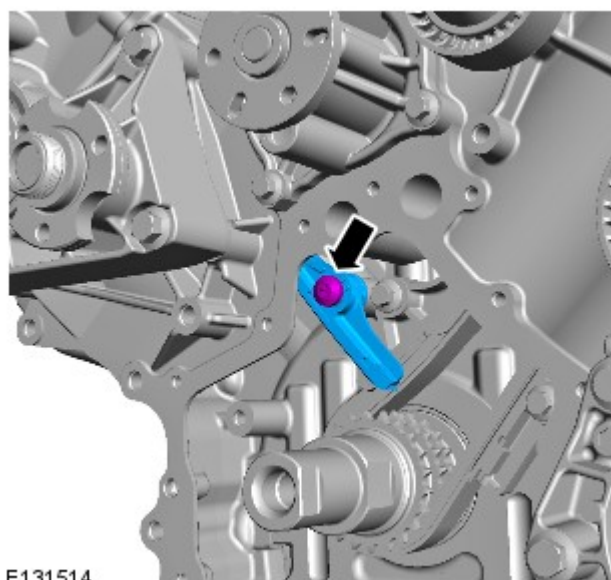
E131516

35.



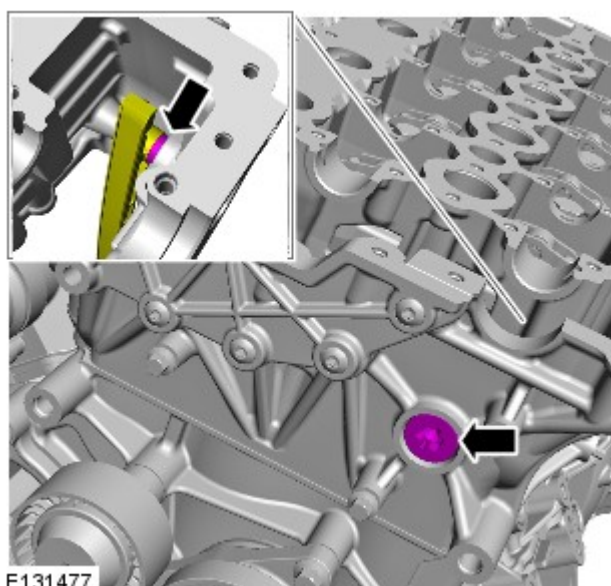
E131515

36.



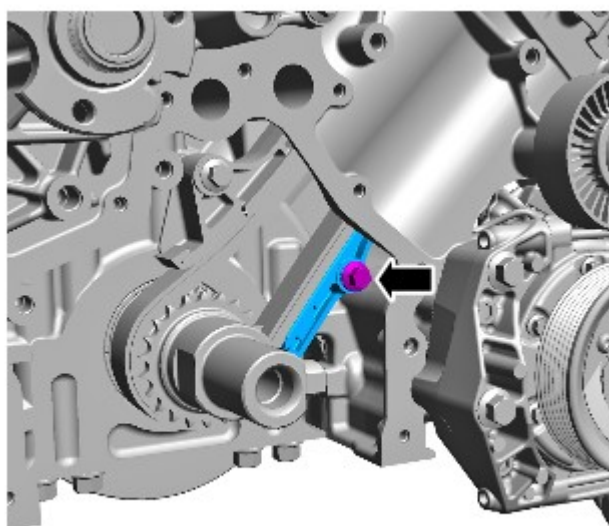
E131514

37.



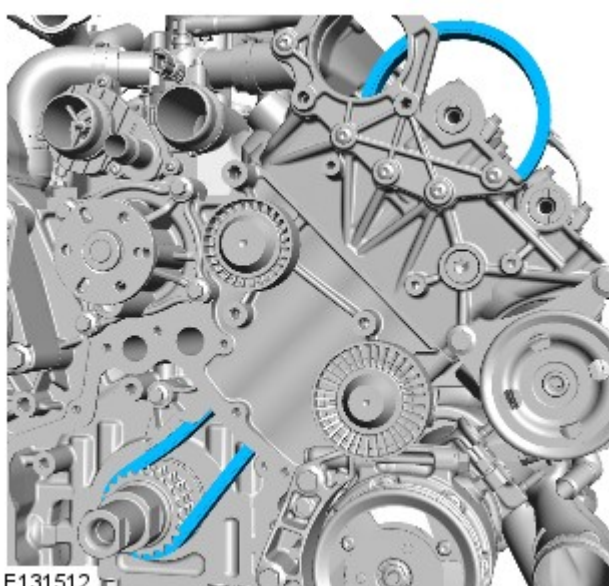
E131477

38.



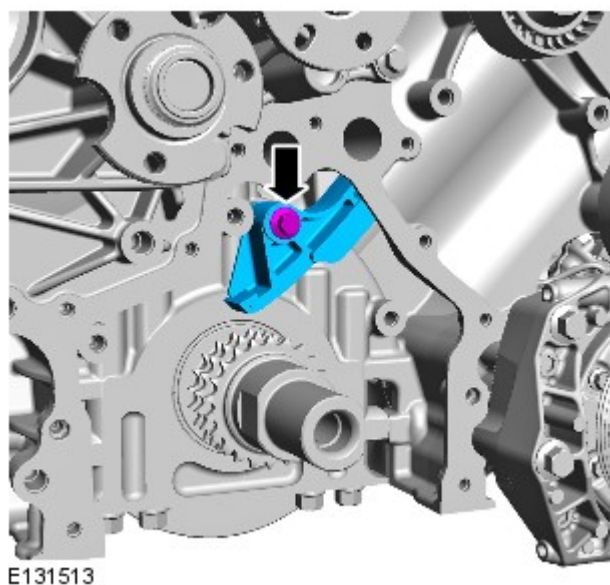
E131511

39.



E131512

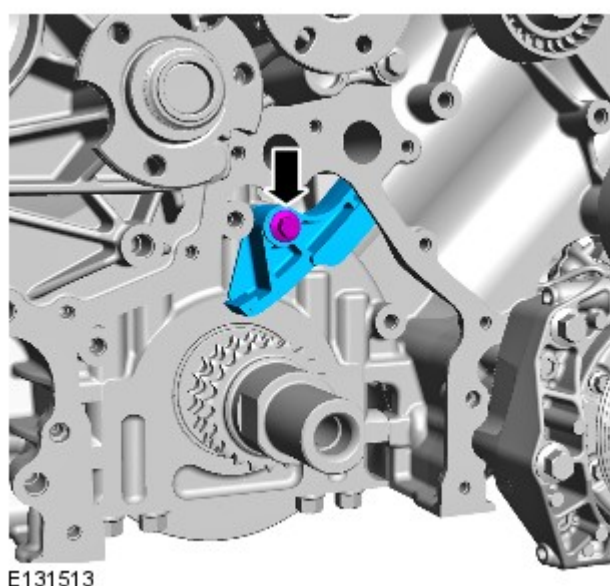
40.



E131513

INSTALLATION

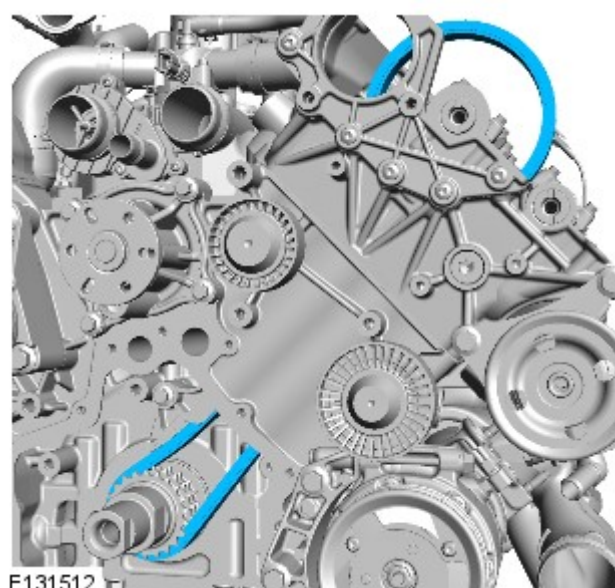
1.



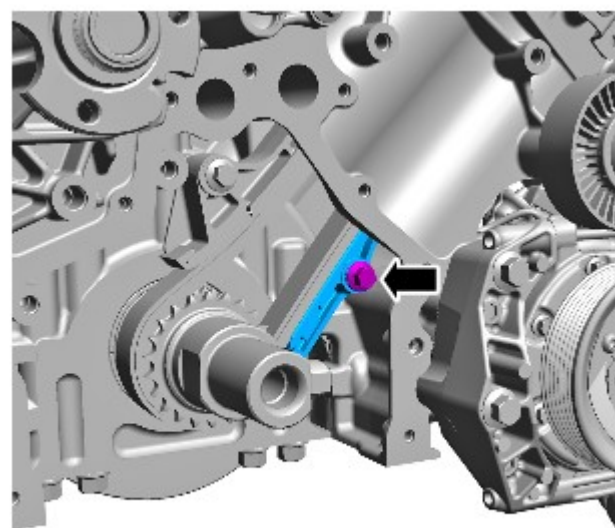
E131513

Torque: 20 Nm

2.

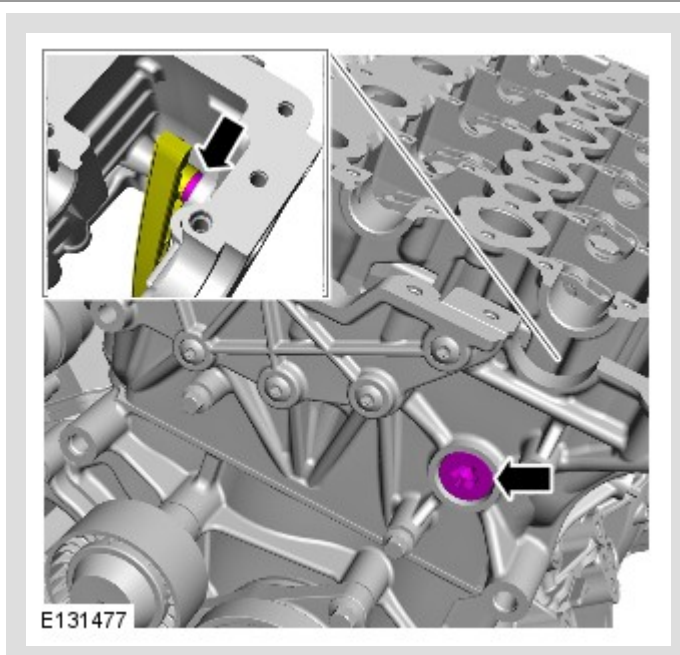


3.



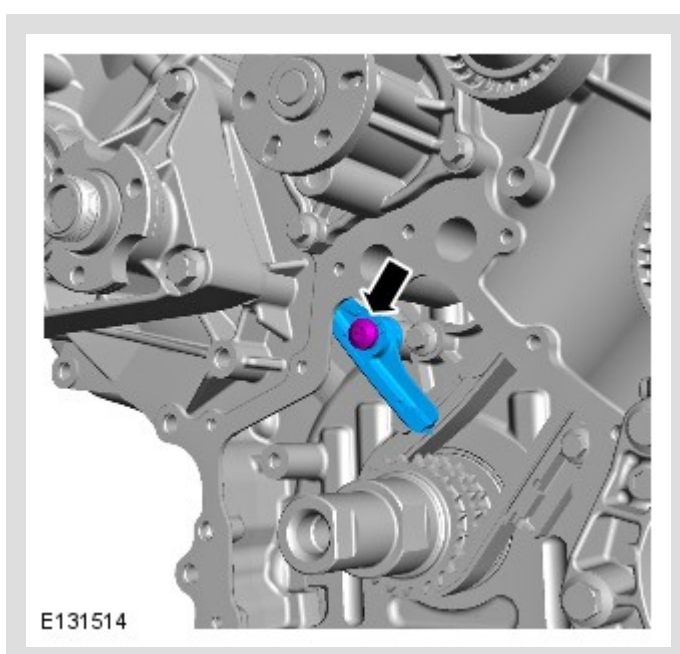
Torque: 20 Nm

4.



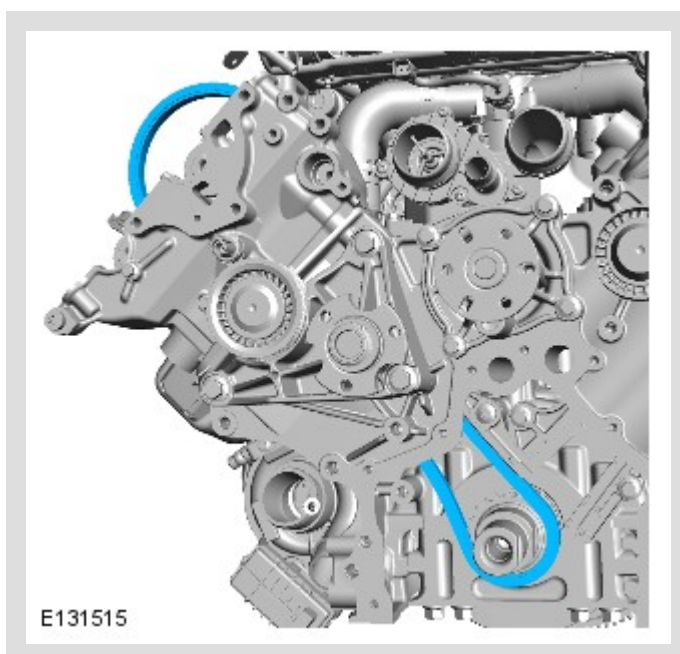
Torque: 27 Nm

5.

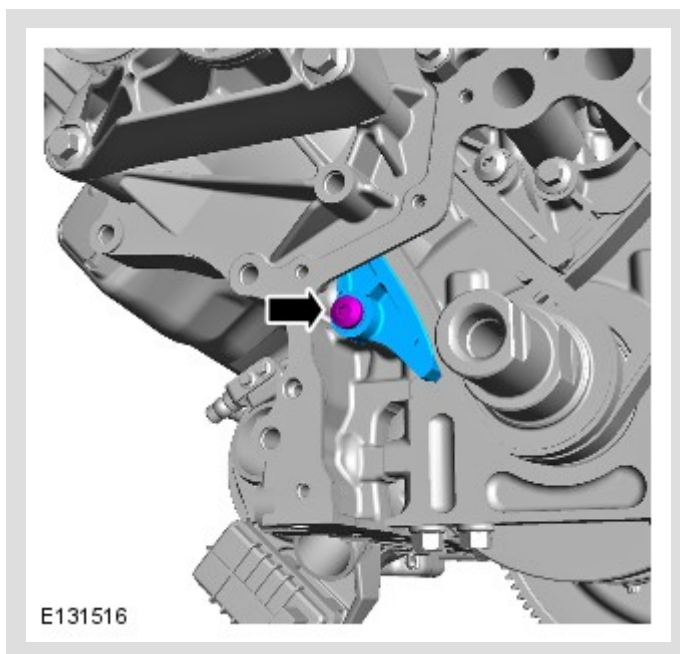


Torque: 20 Nm

6.

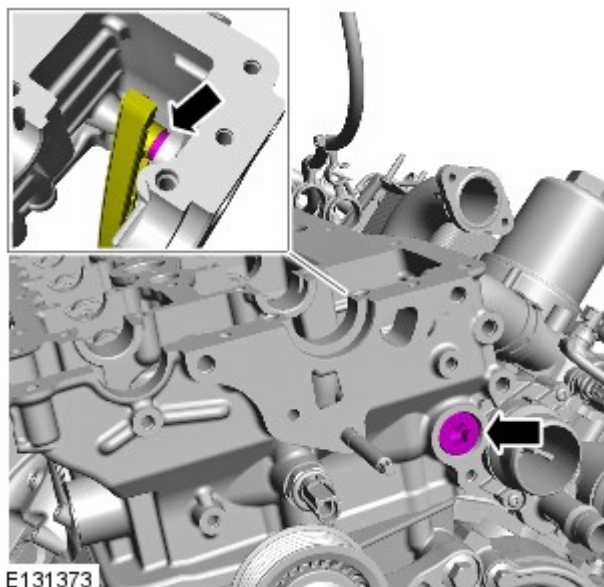


7.



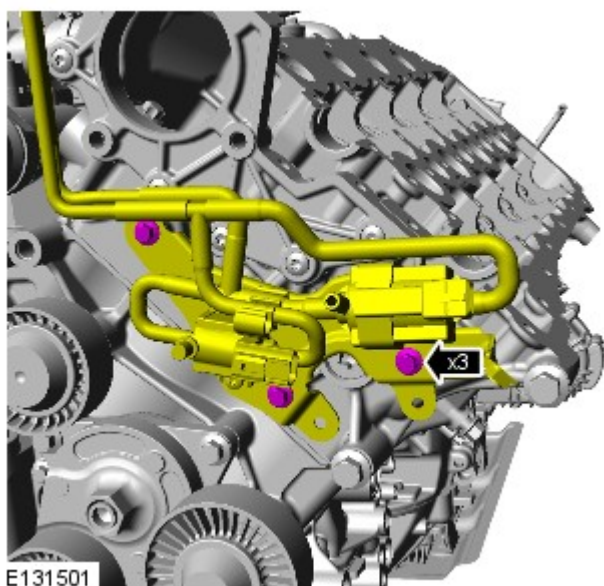
Torque: 20 Nm

8.



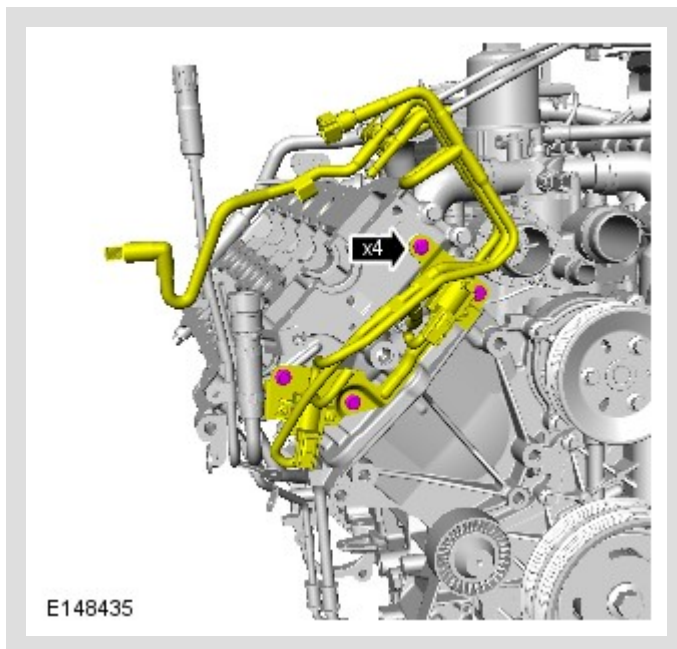
Torque: 27 Nm

9.



Torque: 10 Nm

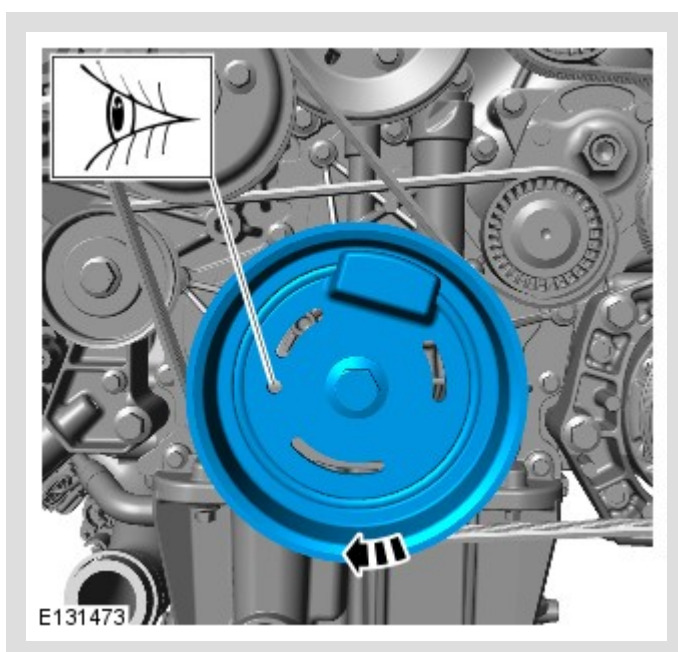
10.



Torque: 10 Nm

-
11. Refer to: [Oil Pump](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).

12.



Rotate the engine clockwise until the crankshaft contacts the timing pin. Do not attempt to rotate the engine counter clockwise.

13.

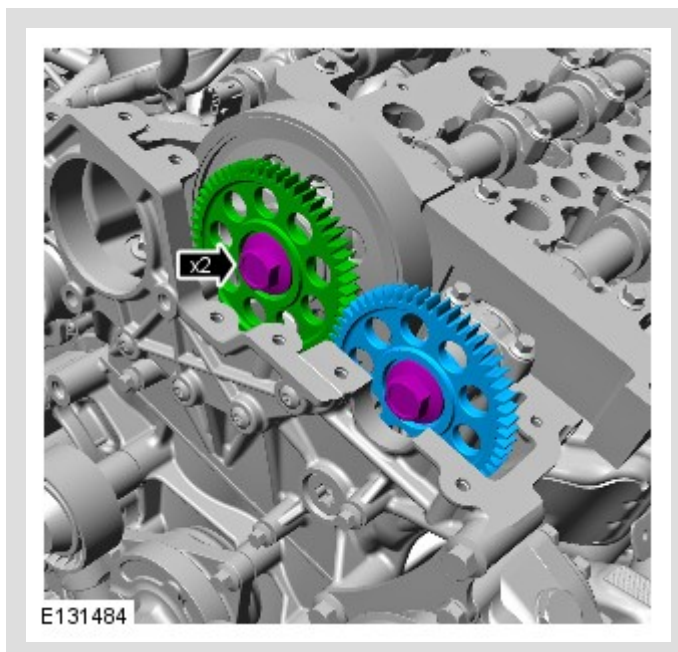


14.



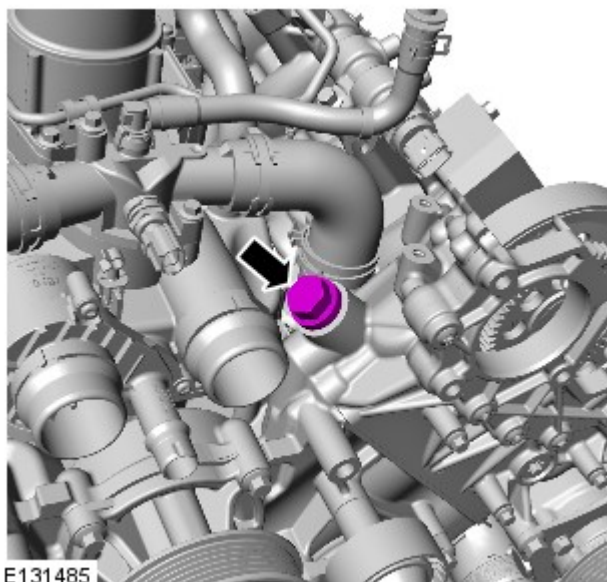
CAUTIONS:

- Make sure that new bolts are installed.
- Only tighten the bolts finger-tight at this stage.



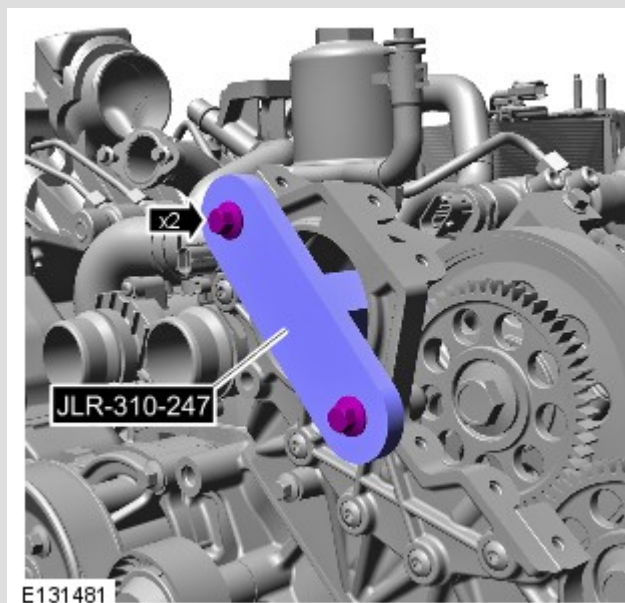
Renew Part: *Camshaft sprocket bolt* : 4 .

15.



Torque: 65 Nm

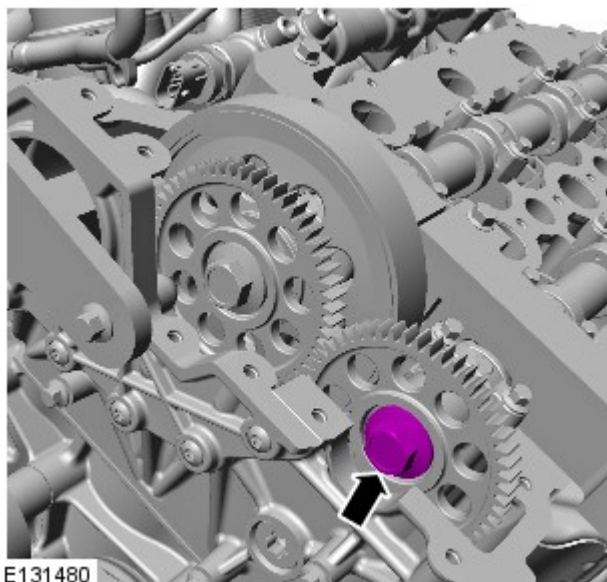
16.



Special Tool(s): [JLR-310-247](#)

Torque: 10 Nm

17.

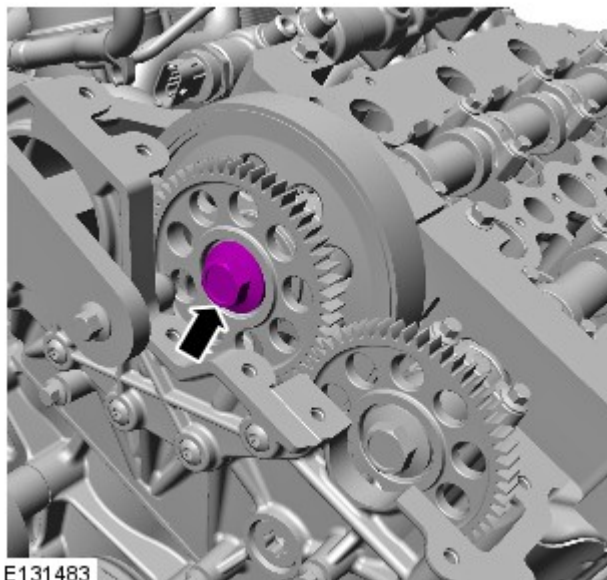


Torque:

Stage 1: **80 Nm**

Stage 2: **80°**

18.

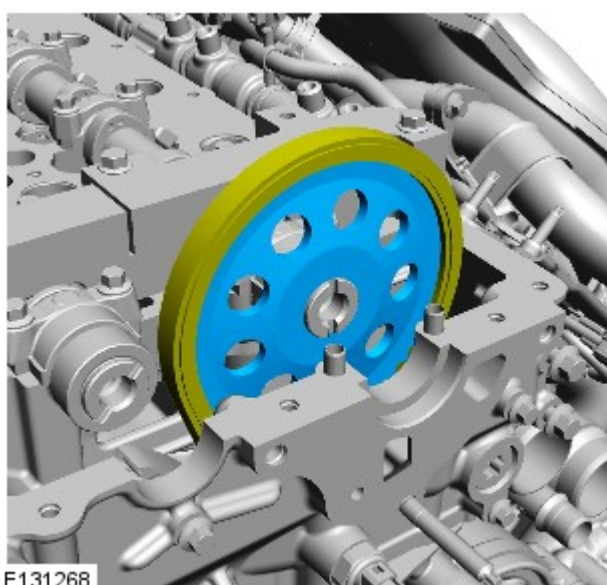


Torque:

Stage 1: 80 Nm

Stage 2: 80°

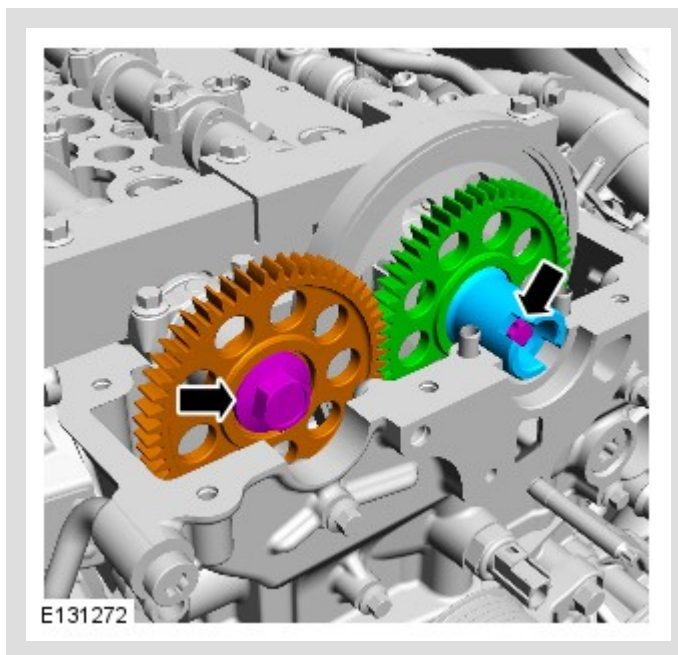
19.



20.

ⓘ CAUTIONS:

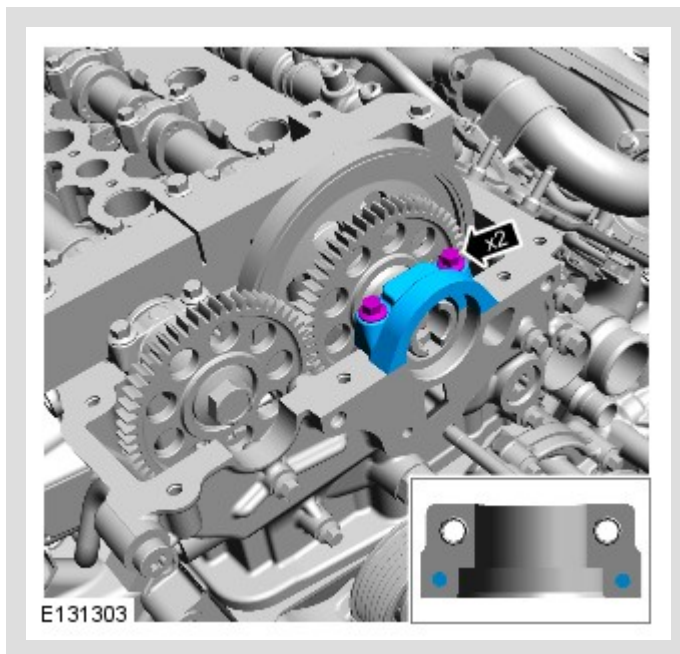
- Make sure that new bolts are installed.
- Only tighten the bolts finger-tight at this stage.



21.

ⓘ CAUTION:

Make sure that the mating faces are clean and free of foreign material.

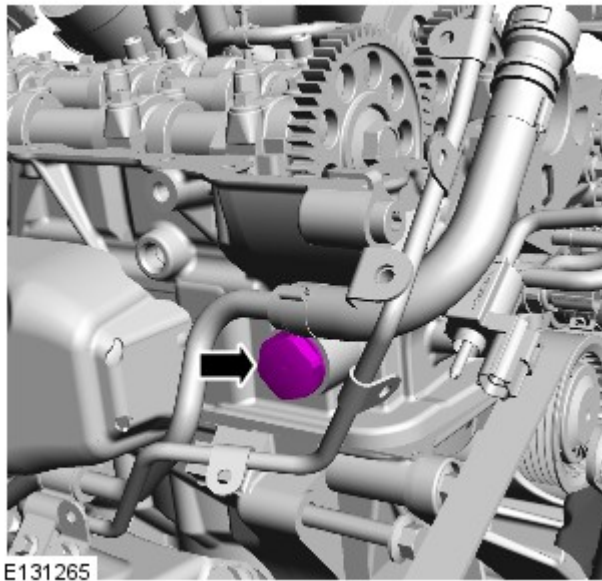


Apply a 3.5 mm - 4 mm circle of loctite 518 sealant in the two areas shown.

Torque:

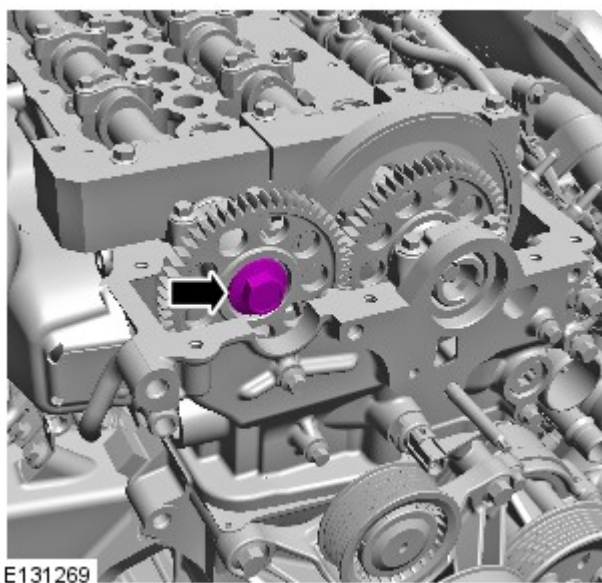
Stage 1: **5 Nm**

Stage 2: **10 Nm**



Torque: 65 Nm

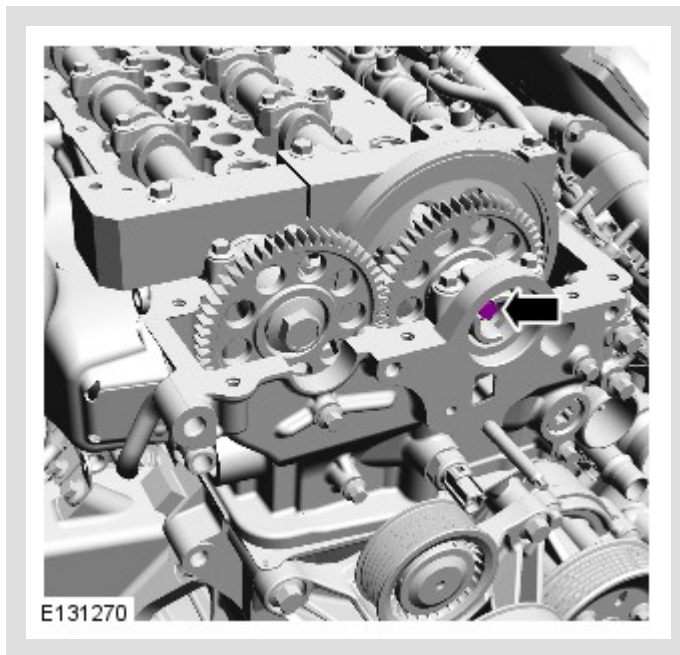
23.



Torque:

Stage 1: 80 Nm

24.

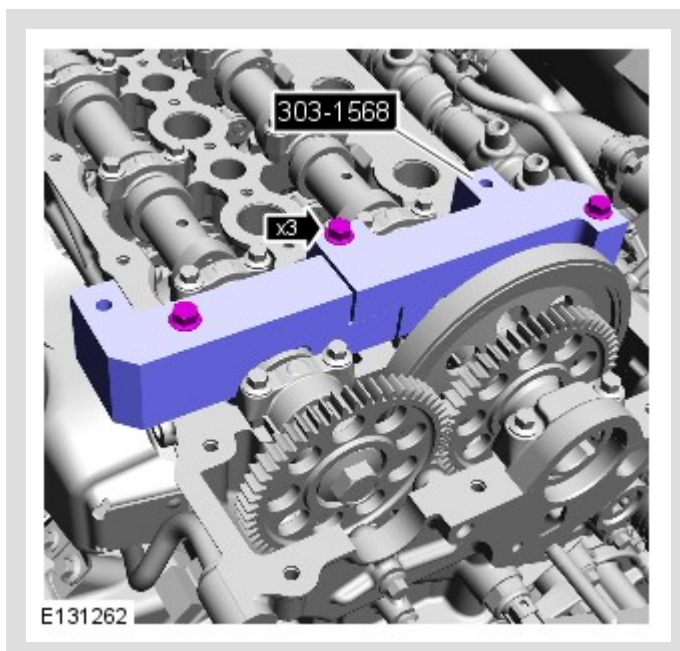


Torque:

Stage 1: 80 Nm

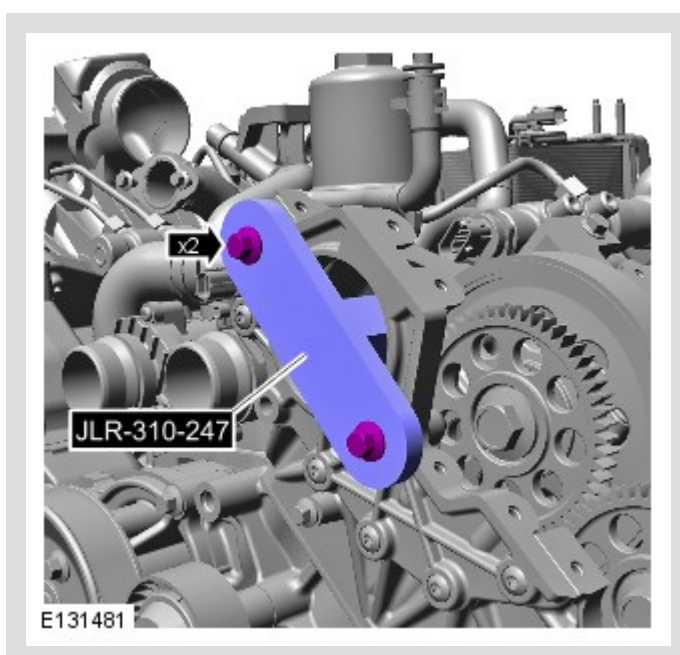
Stage 2: 80°

25.



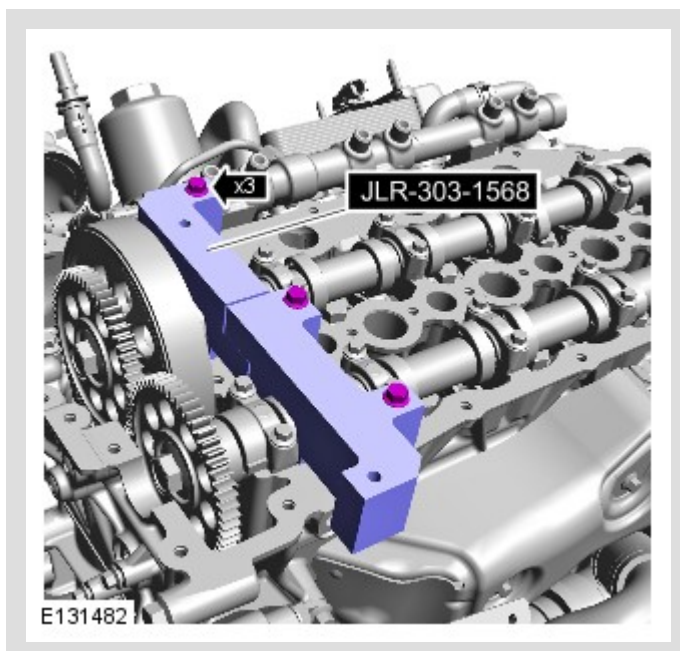
Remove the special tool.

26.



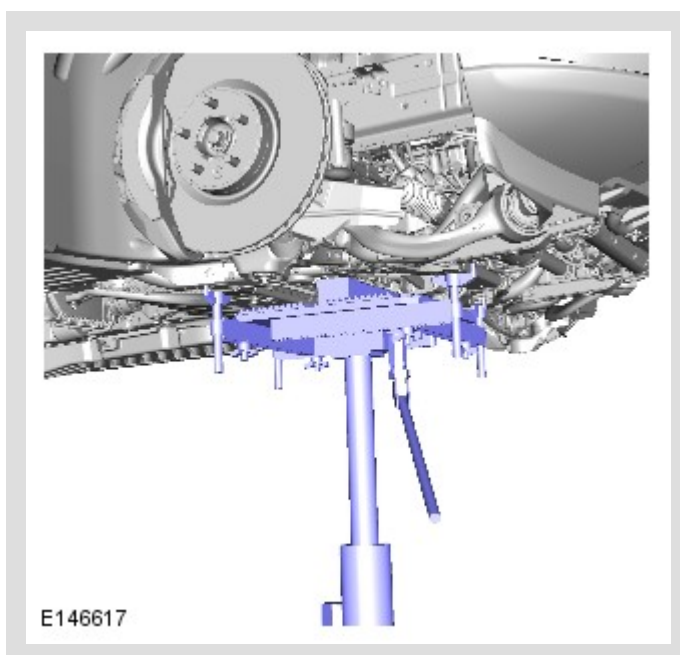
Remove the special tool.

27.



Remove the special tool.

28.

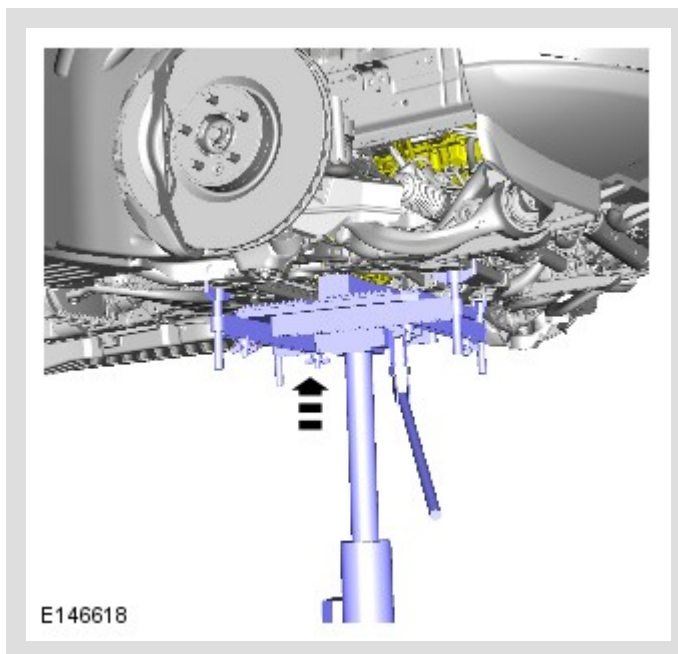


29.

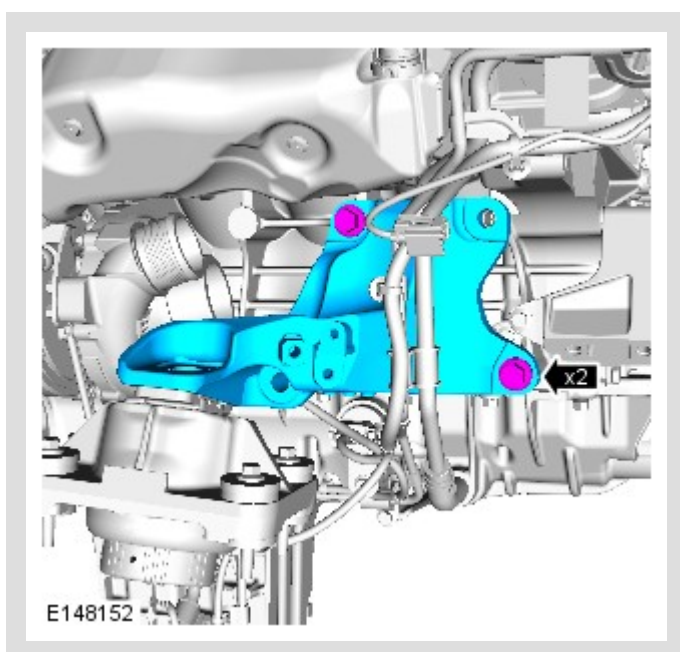


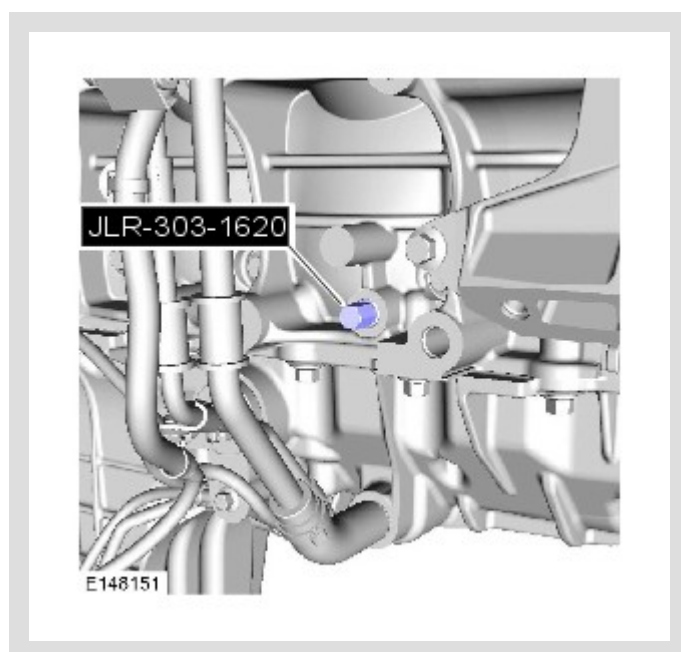
CAUTION:

Use a wooden block to protect the oil pan when supporting the engine.

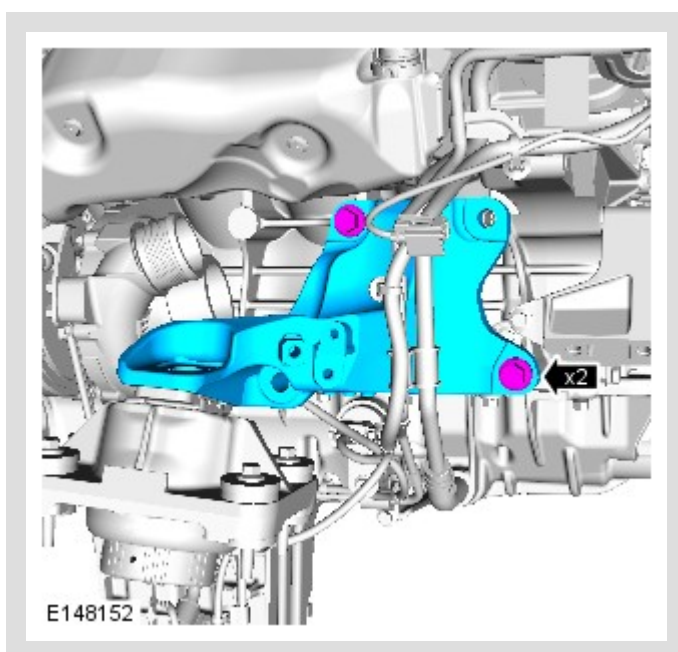


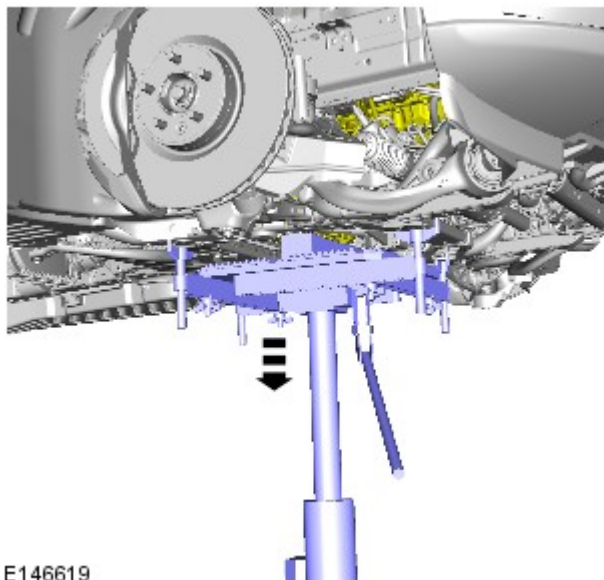
Using a suitable hydraulic jack, raise and support the engine.



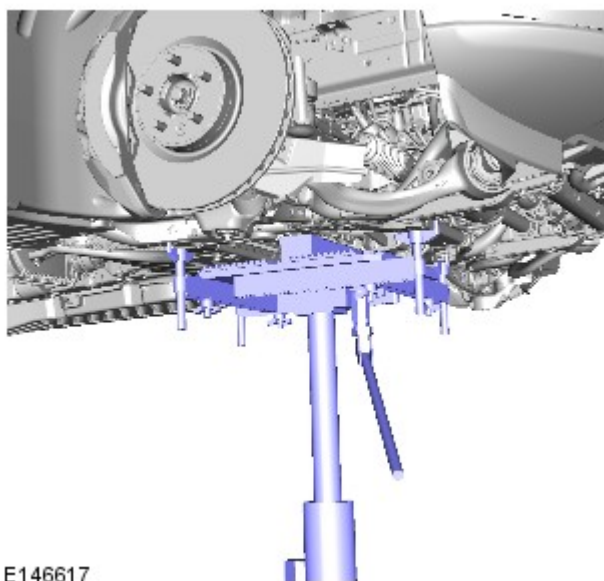


Special Tool(s): [JLR-303-1620](#)



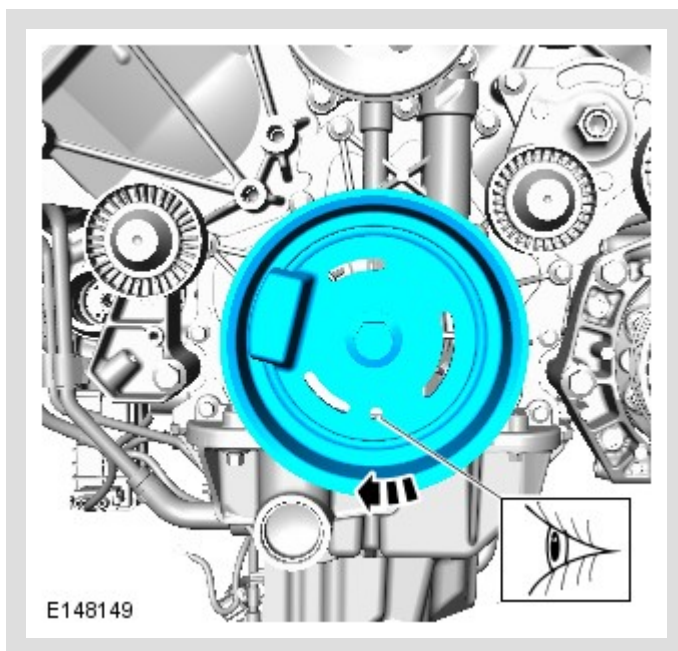


34.

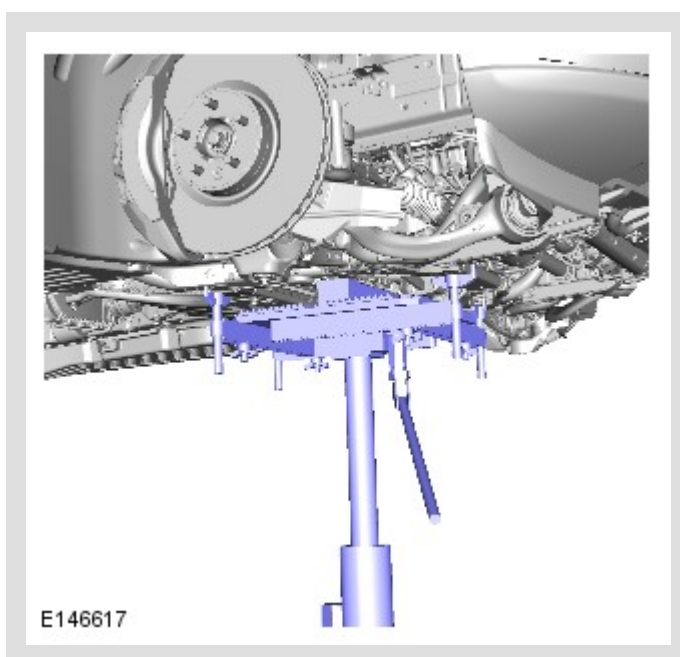


35.

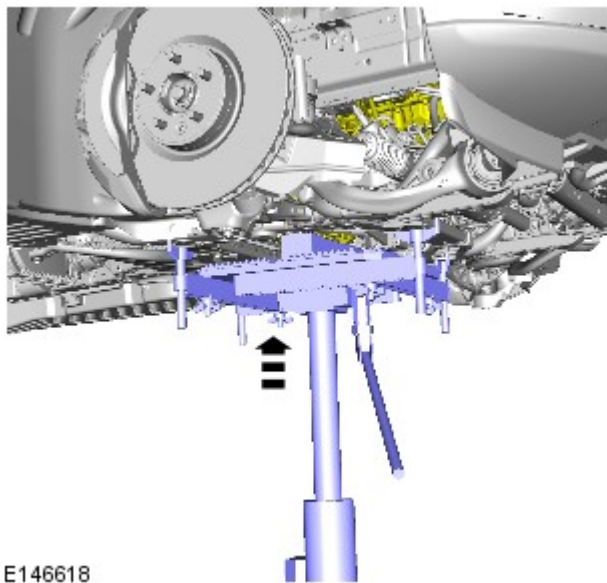
Rotate the engine two complete turns clockwise.

**CAUTION:**

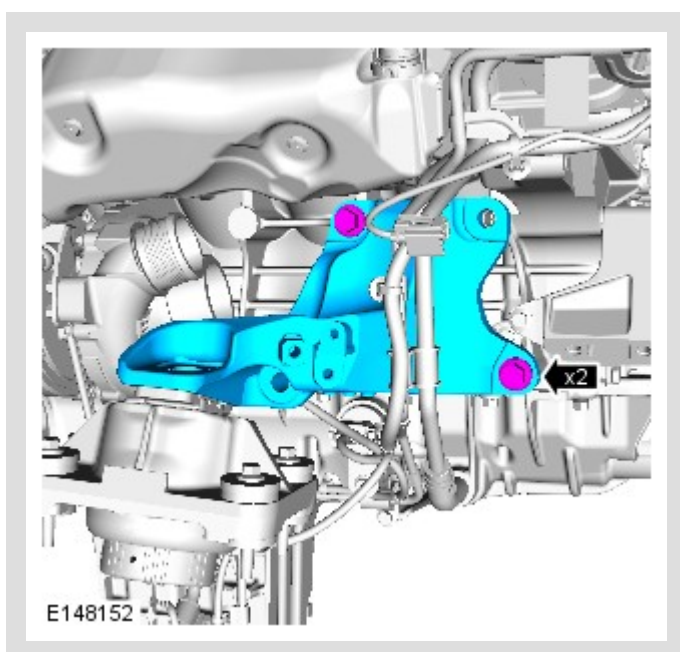
Use a wooden block to protect the oil pan when supporting the engine.

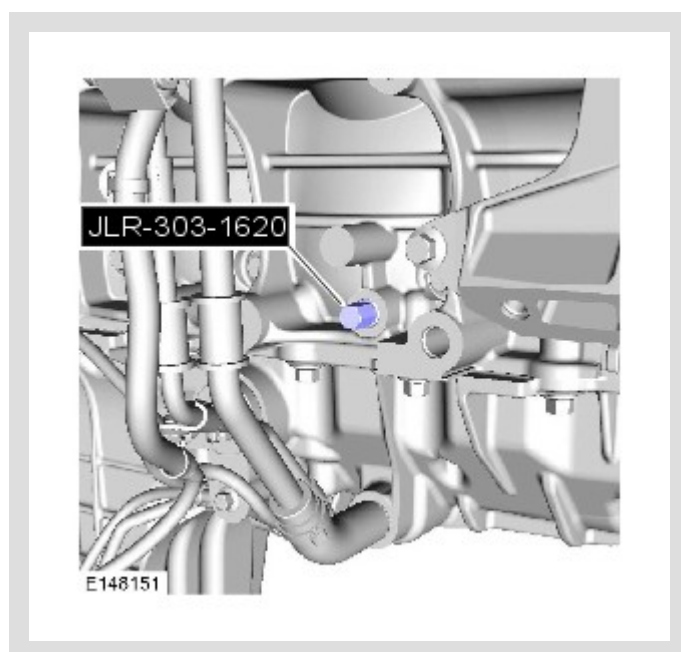


Using a suitable hydraulic jack, raise and support the engine.

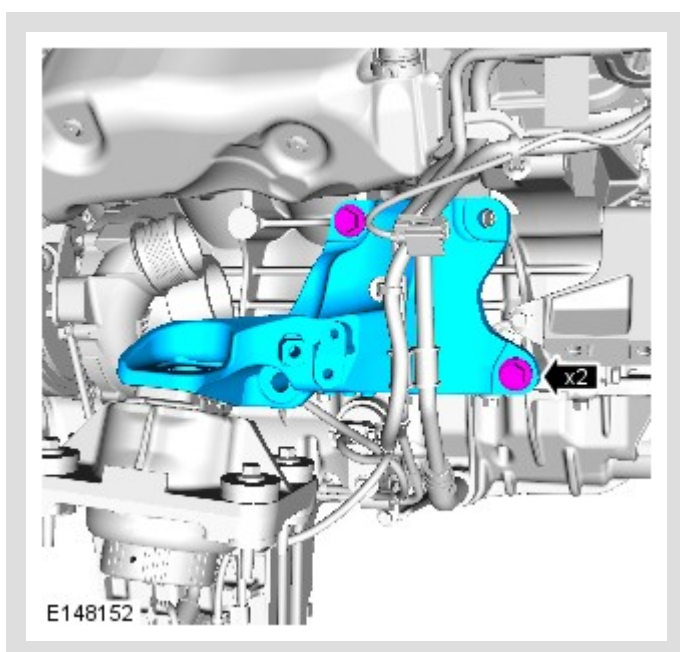


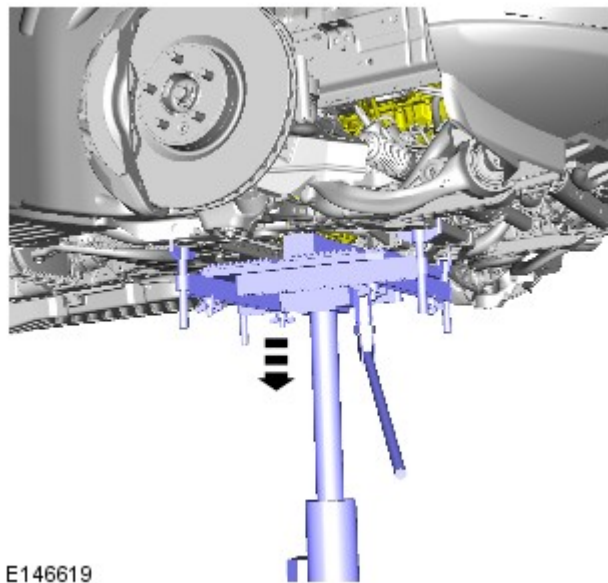
E146618



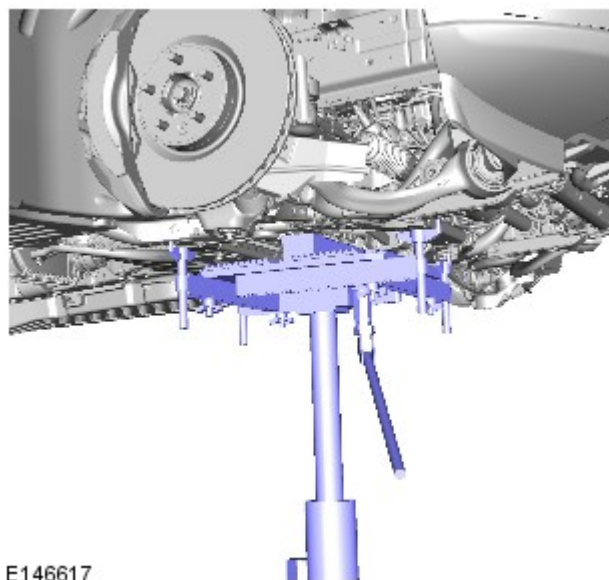


Special Tool(s): [JLR-303-1620](#)

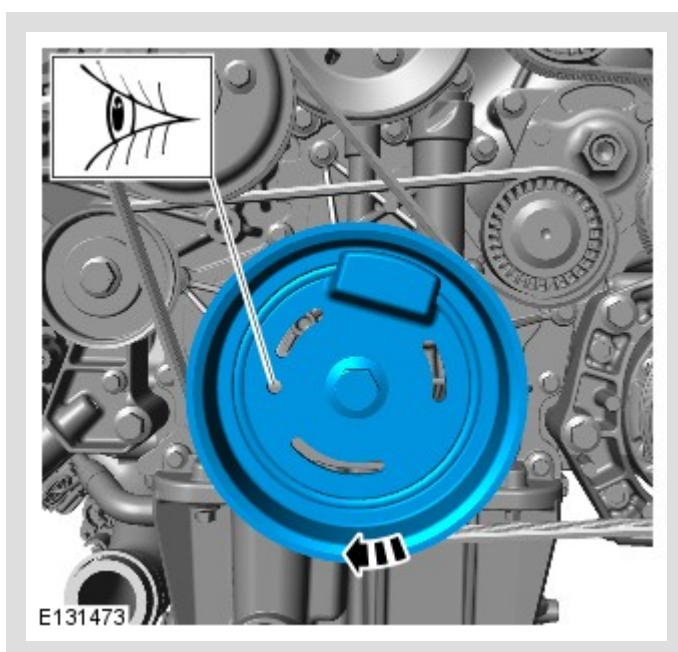




E146619



E146617

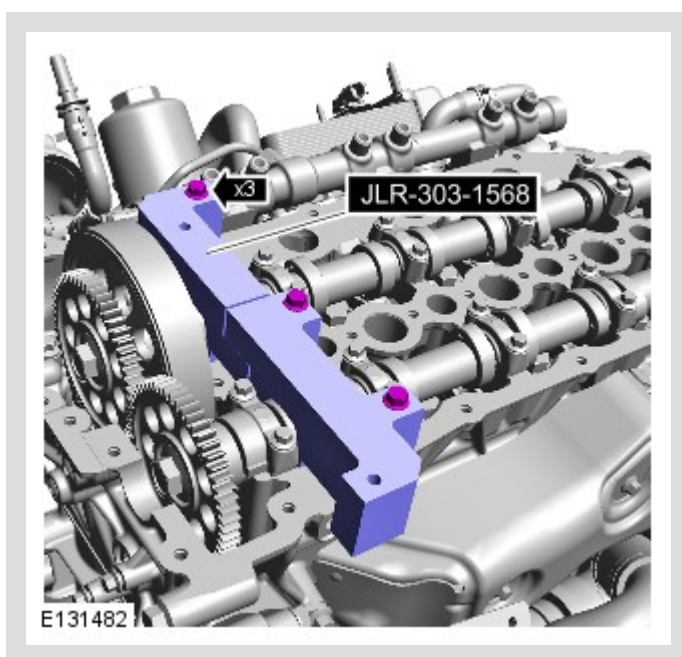


Rotate the engine clockwise until the crankshaft contacts the timing pin. Do not attempt to rotate the engine counter clockwise.

45.

⚠ CAUTION:

If the special tool cannot be installed, the timing chain installation steps must be repeated.



Install the special tool.

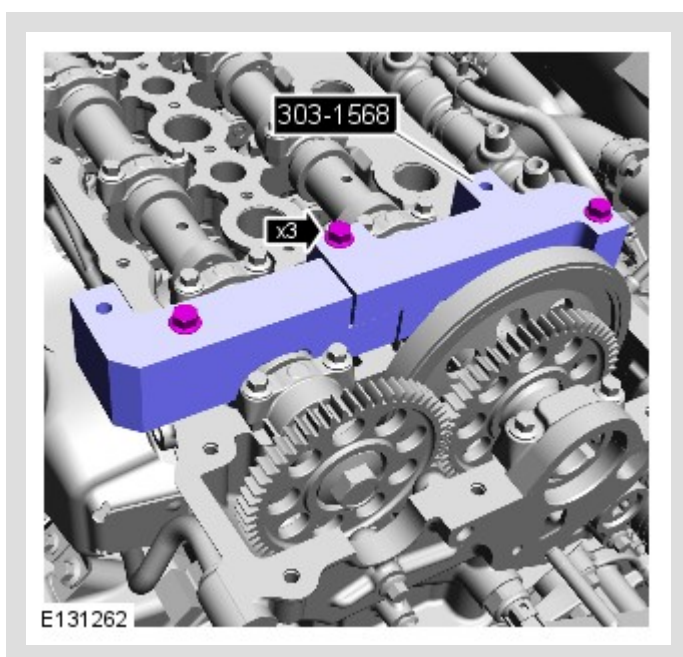
Special Tool(s): [JLR-303-1568](#)

Torque: 10 Nm

46.

⚠ CAUTION:

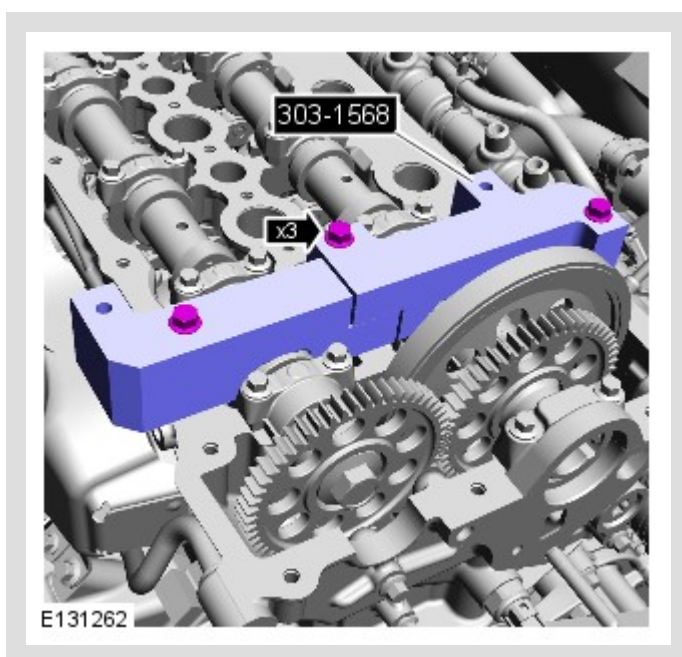
If the special tool cannot be installed, the timing chain installation steps must be repeated.



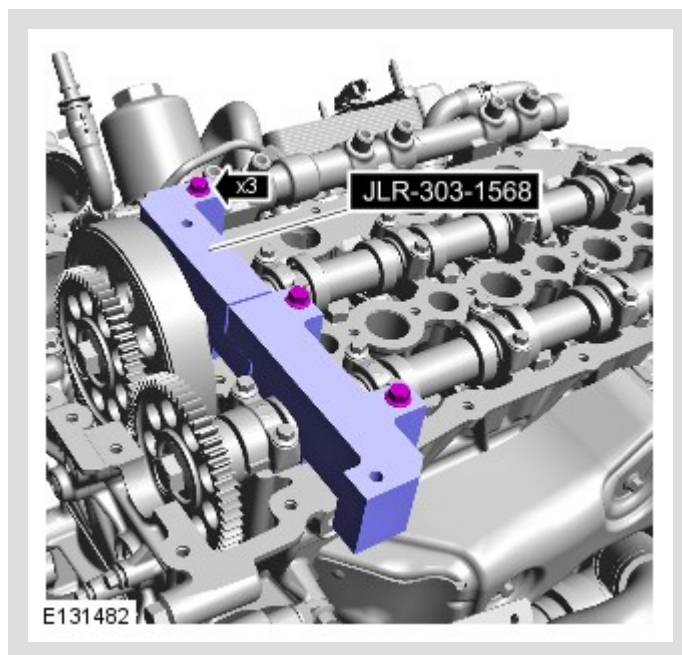
Install the special tool.

Special Tool(s): [JLR-303-1568](#)

Torque: 10 Nm

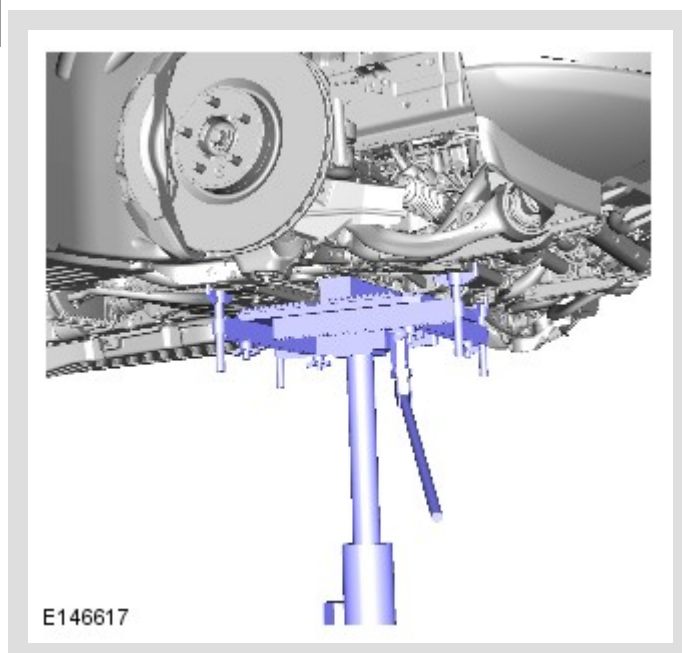


Remove the special tool.

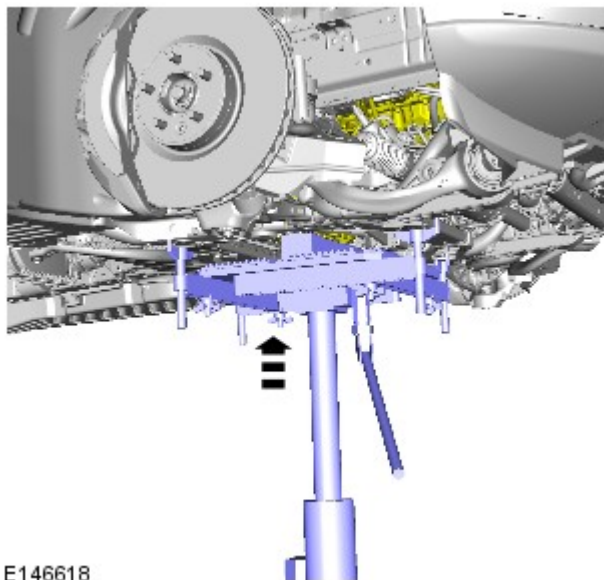


Remove the special tool.

49.

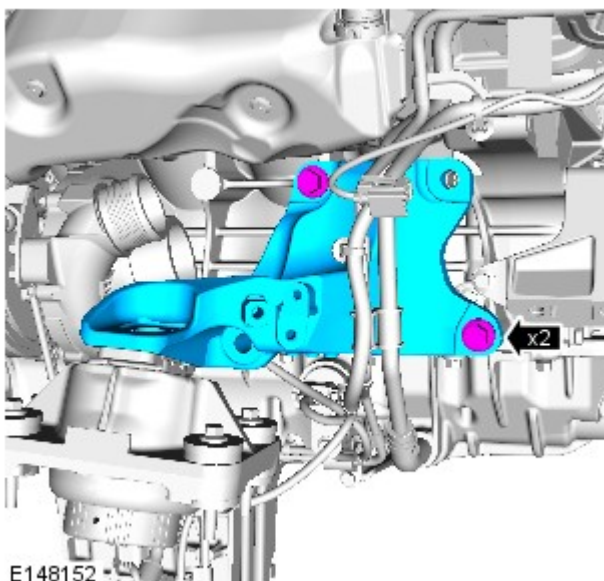


50.



E146618

51.



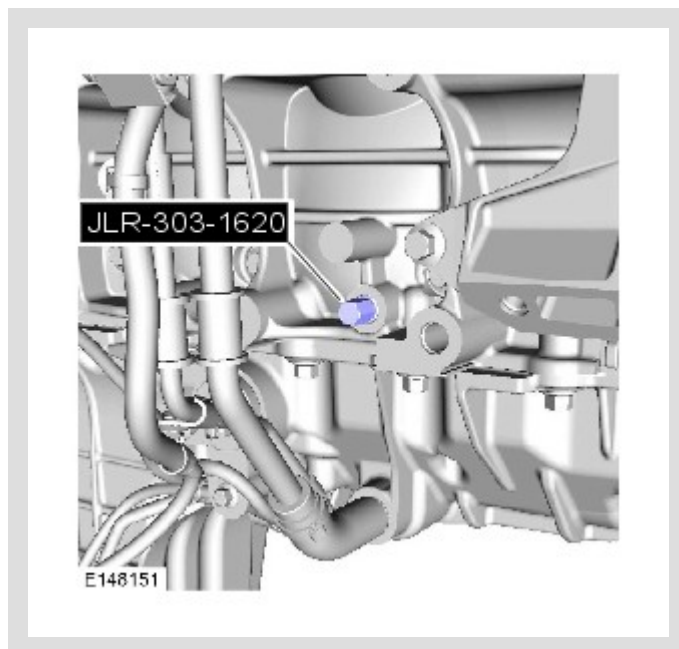
E148152

52.



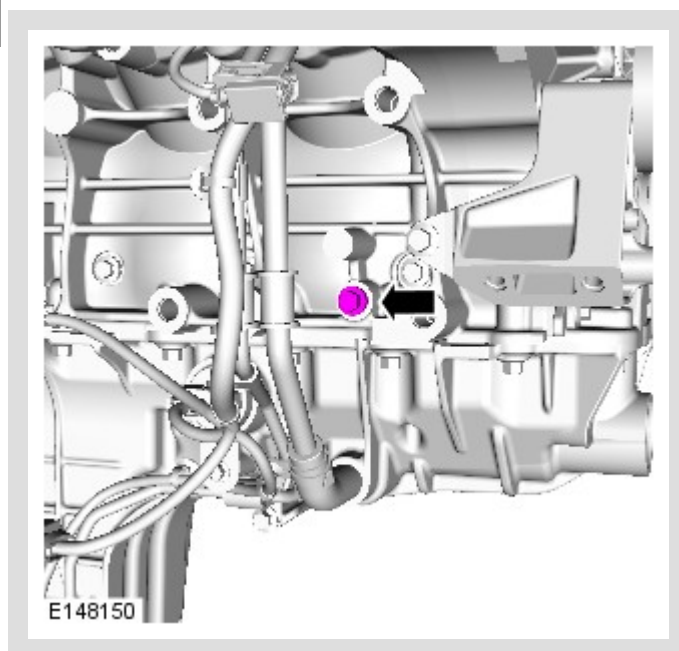
NOTE:

Remove the special tool.



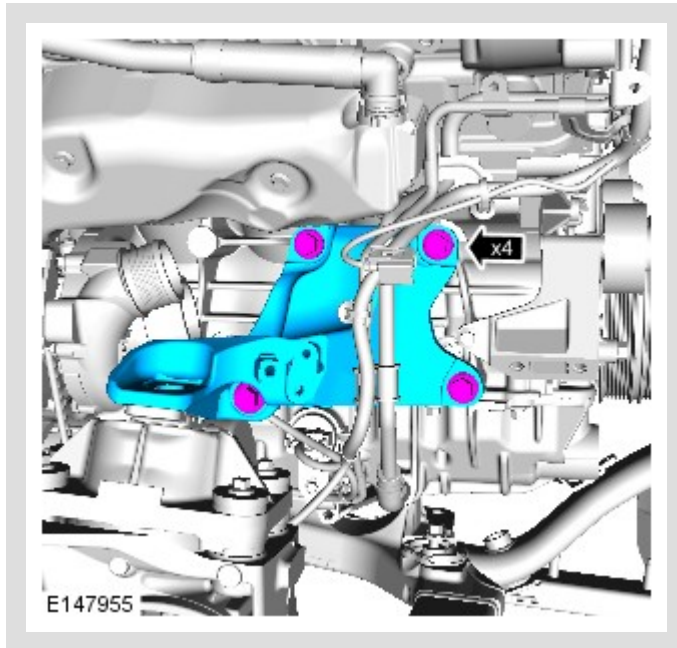
Special Tool(s): [JLR-303-1620](#)

53.



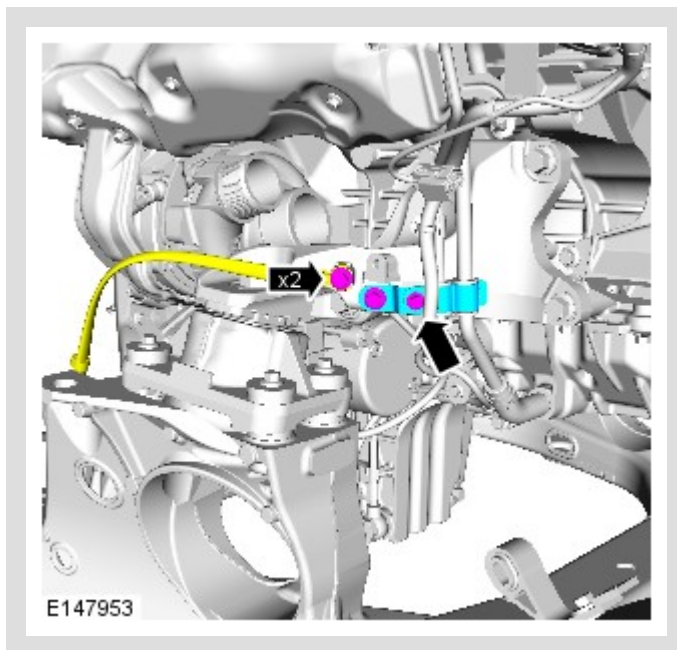
Torque: 23 Nm

54.



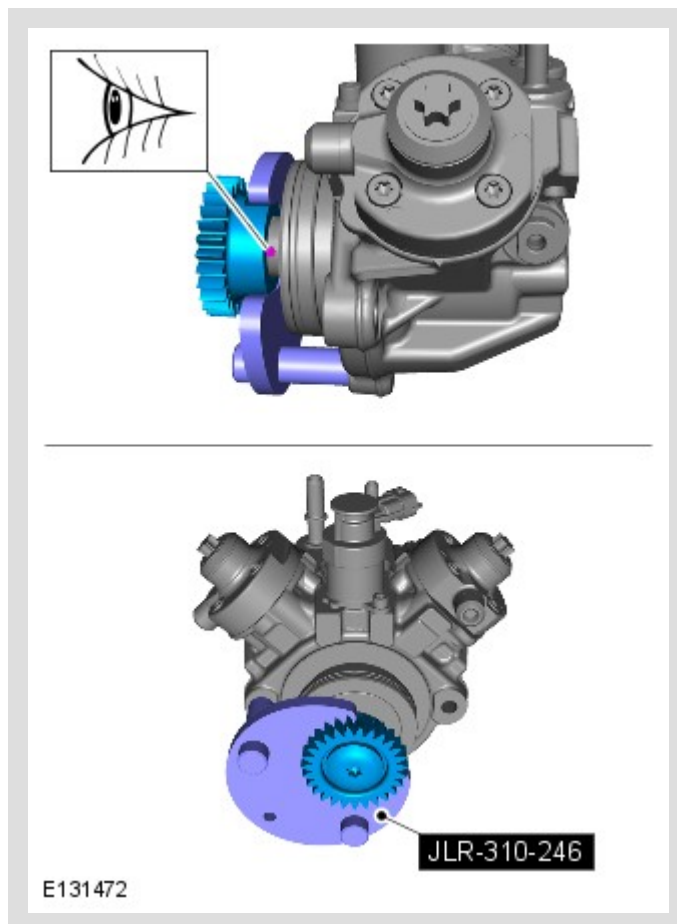
Torque: 119 Nm

55.



Torque:
M8 25 Nm

56.



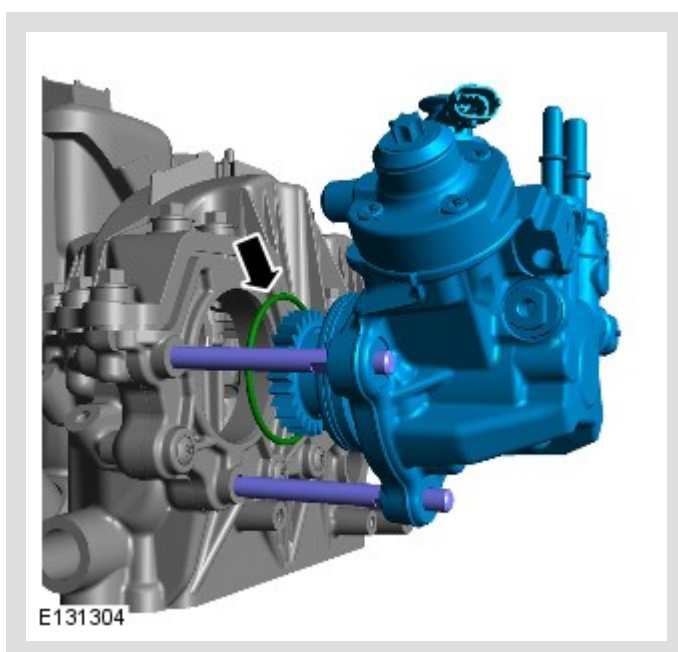
Special Tool(s): [JLR-310-246](#)

57.

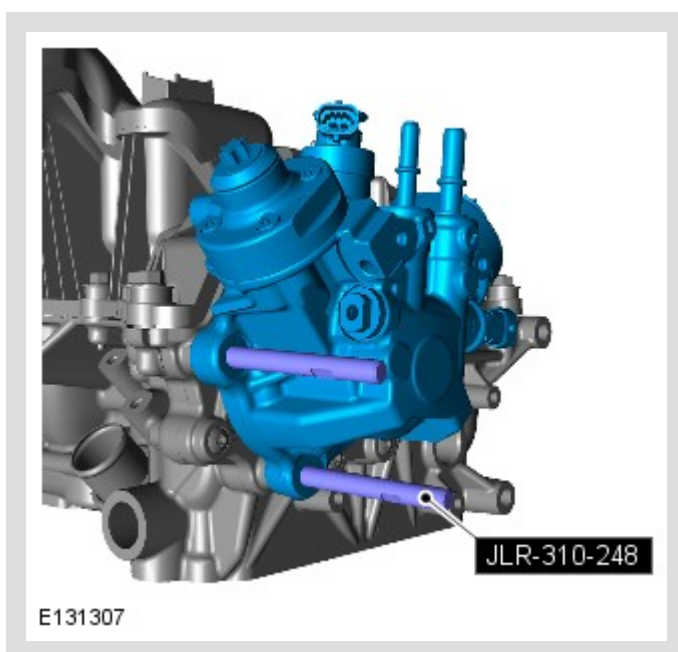


NOTE:

Lubricate the O-ring seal with clean engine oil.



Renew Part: *Fuel pump O-ring seal* : 1 .



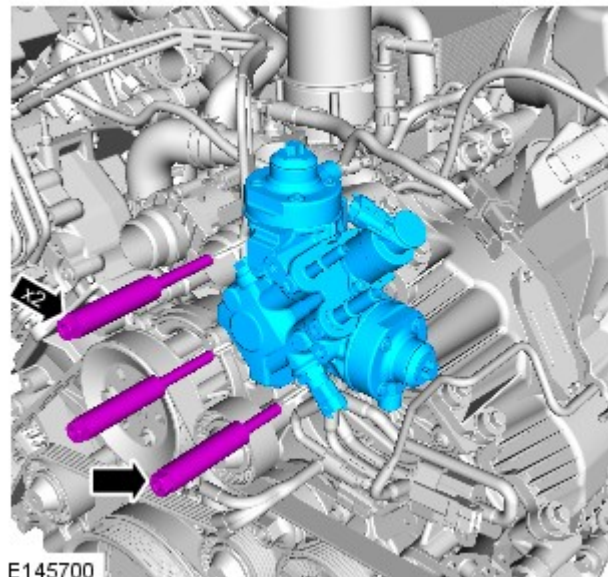
Remove the guide pins.

Special Tool(s): [JLR-310-248](#)

59.

⚠ CAUTIONS:

- Make sure that new bolts are installed.
- Install the bolts in the noted position.



Renew Part: [Fuel Injection Pump retaining bolts](#) : 3 .

Torque: 24 Nm

-
60. Refer to: [Brake Vacuum Pump - TDV8 4.4L Diesel](#) (206-07 Power Brake Actuation, Removal and Installation).
-
61. Refer to: [Generator](#) (414-02B Generator and Regulator - TDV8 4.4L Diesel, Removal and Installation).
-
62. Refer to: [Left Valve Cover](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).
-
63. Refer to: [Right Valve Cover](#) (303-01E Engine - TDV8 4.4L Diesel, Removal and Installation).
-
64. Connect the battery ground cable.
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).